DESIGN AND ANALYSIS OF DEEP BEAM IN ABAQUS

REPORT OF MINI PROJECT - II

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

In

CIVIL ENGINEERING

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DECLARATION

We hereby declare that the Report of the MINI Project- II work entitled "Design and Analysis of Deep Beam in Abaqus" which is being submitted to the National Institute of Technology Karnataka, Surathkal in the award of the Degree of Bachelor of Technology in the department of Civil Engineering, is a bonafide report of the work carried out by us. The material contained in this Project Work Report has not been submitted to any University or Institution for the award of any degree.

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ABSTRACT

Deep beams are structural elements characterized by a small span-to-depth ratio, where conventional flexural theory is not applicable and load transfer primarily occurs through arching action. The behaviour of deep beams under concentrated point loads is complex due to non-linear stress distribution and the presence of discontinuity regions. Accurate analysis and design of such elements require advanced approaches that account for these characteristics.

In the present study, a reinforced concrete deep beam is designed and analysed under point loading conditions using provisions based on the American Concrete Institute (ACI 318). The influence of span-to-depth (L/D) ratio on the structural behaviour is investigated by considering a range of geometric configurations. Finite element analysis is performed using ABAQUS to simulate the response of the deep beam and to capture detailed stress distribution and deformation behaviour under loading.

The study also includes seismic analysis to evaluate the response of the deep beam under dynamic conditions. Reinforcement detailing, including vertical and horizontal reinforcement, is incorporated to ensure adequate strength and stability, with considerations for improving performance through additional reinforcement strategies. The work aims to develop a comprehensive understanding of the behaviour of deep beams under both static and dynamic loading conditions using analytical and numerical approaches.

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1. INTRODUCTION

1.1.Deep Beam

A deep beam is a structural member characterized by a relatively small span-to-depth ratio, resulting in a significant deviation from the assumptions of classical beam theory. According to the provisions of IS 456:2000, a beam may be classified as a deep beam when the ratio of effective span to overall depth is less than or equal to 2.0 for simply supported beams and 2.5 for continuous beams. Under such conditions, the assumption that plane sections remain plane after bending is no longer valid, and the strain distribution across the depth becomes non-linear.

Similarly, as per ACI 318, a deep beam is defined as a structural element with a small span-to-depth ratio in which the load transfer mechanism is dominated by direct compression between the load and the supports rather than flexural action. Typically, members with a clear span-to-depth ratio less than or equal to 4.0 are considered deep beams. In such members, the internal stress distribution is better represented by a system of compression struts and tension ties instead of the conventional bending theory.

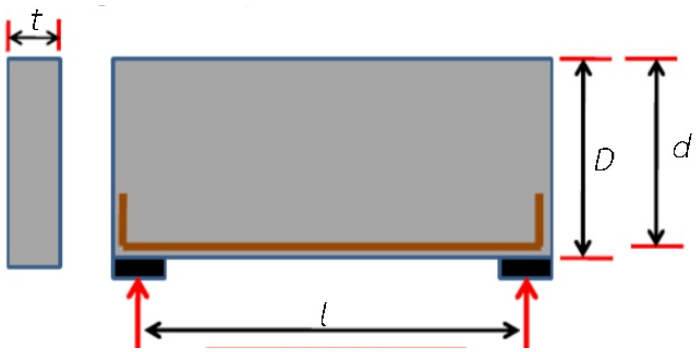


Figure 1:Deep Beam

The structural behaviour of deep beams is fundamentally different from that of slender beams. Due to their geometry, shear deformations become significant, and the load is transmitted directly from the point of application to the supports through inclined compression paths. This results in a truss-like action within the member, where concrete primarily resists compressive forces through struts, and reinforcement carries tensile forces through ties. Consequently, the design of deep beams requires special consideration of shear strength, anchorage, and detailing to ensure safe load transfer.

In modern practice, the analysis and design of deep beams are often carried out using the strut-and-tie method, which provides a rational representation of the internal force flow. This approach is particularly emphasized in ACI provisions and is increasingly adopted in design standards due to its ability to accurately model discontinuity regions. Deep beams are commonly used in structural applications such as transfer girders, pile caps, and bridge pier caps, where concentrated loads and short spans necessitate efficient load transfer mechanisms.

1.2.Difference Between Deep Beam v/s Shallow Deep

Structural beams are generally classified as deep beams or shallow (slender) beams based on their span-to-depth ratio and the nature of stress distribution. This classification is important because it directly influences the analysis assumptions and design methodology. According to IS 456:2000 and ACI 318, deep beams are identified when the span-to-depth ratio is relatively small, whereas shallow beams possess a larger span relative to their depth.

In shallow beams, the assumptions of classical bending theory are valid. Plane sections remain plane after bending, and the strain distribution across the depth is linear. As a result, the behaviour is primarily governed by flexure, with shear playing a secondary role. These beams are typically designed using standard bending theory and conventional shear design equations. In contrast, deep beams do not satisfy these assumptions. Due to their small span-to-depth ratio, the strain distribution becomes non-linear, and significant shear deformations occur. The load is transferred more directly from the point of application to the supports through compression paths, resulting in a fundamentally different internal force mechanism.

Another key distinction lies in the mode of load transfer. Shallow beams resist loads mainly through bending moments, with tensile stresses taken by longitudinal reinforcement and compressive stresses by concrete. However, deep beams behave more like a truss system, where loads are carried through a combination of compression struts in concrete and tension ties in reinforcement. This necessitates the use of the strut-and-tie method for accurate analysis and design, especially as recommended in modern provisions such as those in ACI 318.

From a geometric perspective, shallow beams typically have a span-to-depth ratio greater than about 2.5 to 4, depending on support conditions, whereas deep beams fall below these limits ($\leq$ 2.0 for simply supported beams as per IS 456, and ≤ 4.0 as per ACI guidelines). Due to this, deep beams exhibit higher shear stresses and require special attention to detailing, anchorage, and confinement reinforcement.

In terms of applications, shallow beams are commonly used in typical building frames where spans are relatively long and loads are moderate. Deep beams, on the other hand, are used in situations involving heavy concentrated loads and short spans, such as transfer girders, pile caps, and bridge pier caps. Their ability to transfer loads efficiently over short distances makes them particularly suitable for such structural configurations.

2. MODELLING IN ABAQUS SOFTWARE

2.1. MODULES OF ABAQUS

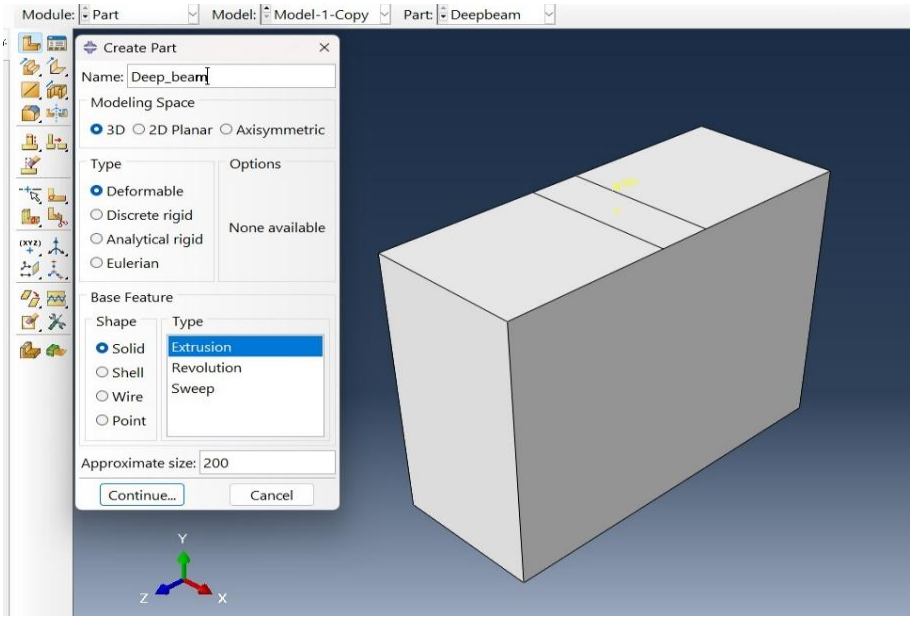


Figure 2:View of Part Module

The Part Module serves as the primary environment for geometric definition, where components are established as 3D Deformable bodies to accurately capture structural behavior. The modelling process begins in the Sketcher, utilizing an Approximate Size grid to draw the precise 2D cross-section of the beam, which is then converted into a 3D volume via the Solid Extrusion base feature. To facilitate complex analysis, Datum Tools are used to create reference points and planes, allowing for strategic Partitioning of the part to define specific loading zones and mesh regions. The module also features a comprehensive Model Tree, providing a centralized interface to manage, rename, and edit features like cuts or extrusions. By integrating these sketching and management tools, the module ensures the geometric precision required for various span-to-depth ratios while maintaining a clean parent-part architecture for the rest of the simulation.

Table 1: Part Definitions and Element Types

PART	TYPE	BASE FEATURE
Deep beam	Deformable	Solid Extrusion
Top bar	Deformable	Wire Planar
Bottom bar	Deformable	Wire Planar
Stirrup	Deformable	Wire Planar
Support Plate	Discrete Rigid	Wire Planar

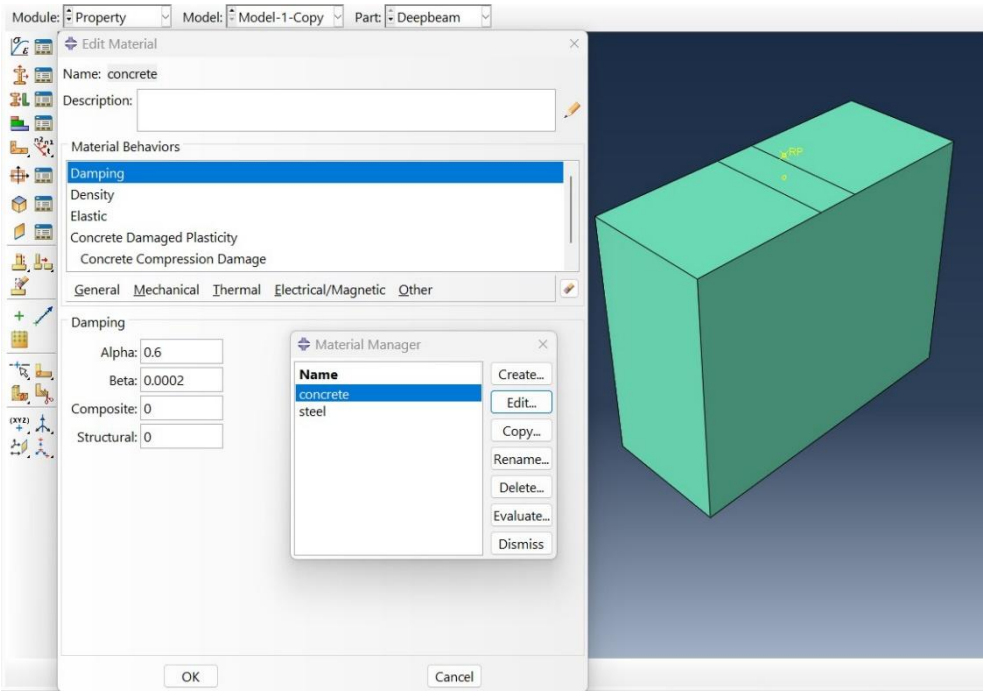


Figure 3:View of Property Module

The Property Module is used to define the physical and mechanical characteristics of the beam through the creation and assignment of complex material models. Using the Material Manager, distinct profiles for "concrete" and "steel" are established, incorporating Elastic properties alongside non-linear behaviors such as Concrete Damaged Plasticity (CDP) to simulate cracking and crushing. The module allows for the input of specific Damping coefficients (alpha = 0.6, beta = 0.0002s) and Density to account for dynamic effects and self-weight during seismic loading. Once materials are defined, Sections are created (e.g., Solid Homogeneous for concrete and Truss/Beam for rebar) and assigned to the respective geometries to link mathematical properties to the physical parts. Finally, the module ensures that the correct material orientation and profile are applied, providing the necessary foundation for the software to calculate the structural response under external forces.

Materials And Its Behaviours:

- Concrete

  Damping: Alpha = 0.6, Beta = 0.0002

  Density = 2.4e -06 tones/mm3

  Elastic: Young's Modulus = 27386 MPa, Poisson's Ratio = 0.2

Table 2:Concrete Damage Plasticity

Dilation angle	Eccentricity	fb0/fc0	K	Viscosity parameter
34	0.1	1.15	0.67	1e-05

Table 3:Concrete Compression Damage

Yield Stress	Inelastic Strain	Damage Parameter	Inelastic Strain
16.8	0	0	0
19.9	8.5E-06	8.5E-06	8.5E-06
22.7	2.4E-05	2.4E-05	2.4E-05
25.1	5E-05	5E-05	5E-05
27.6	9E-05	9E-05	9E-05
29.5	0.00014	0.00014	0.00014
31.2	0.00022	0.00022	0.00022
32.8	0.00031	0.00031	0.00031
33.9	0.00042	0.00042	0.00042
34.5	0.00056	0.00056	0.00056
34.9	0.0007	0.0007	0.0007
34.6	0.00085	0.00085	0.00085
34.1	0.00105	0.00105	0.00105
33.2	0.0013	0.0013	0.0013
31.8	0.0016	0.0016	0.0016
30.2	0.00195	0.00195	0.00195
28.5	0.0023	0.0023	0.0023
26.9	0.00265	0.00265	0.00265

Table 4:Concrete Tension Damage

Yield Stress	Cracking Strain	Damage Parameter	Cracking Strain
4	0	0	0
2.2	0.0003	0.42	0.0003
1.6	0.00055	0.58	0.00055
1.25	0.0008	0.68	0.0008
1.05	0.00105	0.73	0.00105
0.9	0.0013	0.77	0.0013
0.78	0.00155	0.8	0.00155
0.7	0.0018	0.83	0.0018
0.6	0.00265	0.86	0.0021

- Steel

  Damping: Alpha = 0.2, Beta = 5e-05

  Density = 7.85e-06

  Elastic: Young's Modulus = 200000, Poisson's ratio = 0.3

  Plastic: Yield Stress = 500, Plastic Strain = 0

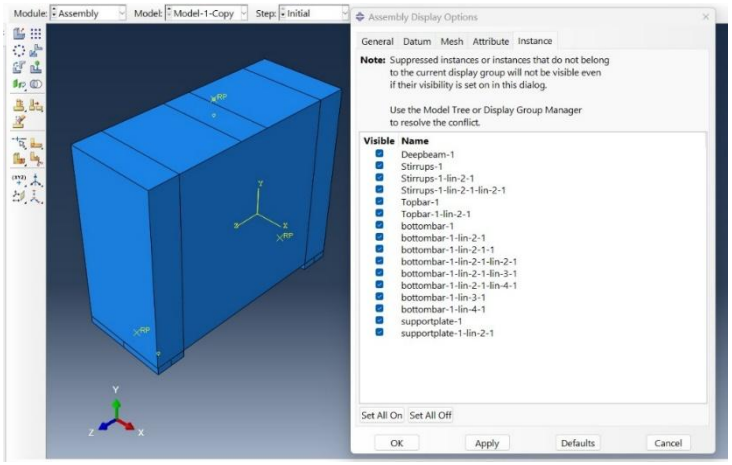


Figure 4:View of Assembly Module

The Assembly Module serves as the centralized environment where individual parts are positioned to create the complete structural system. This module utilizes Instances to bring multiple components, such as the deep beam, reinforcement bars, and support plates, into a single global coordinate system. Using the Translate and Rotate tools, these instances are precisely aligned to ensure correct structural connectivity and overlapping geometries. The interface includes an Assembly Display Options manager that allows users to toggle the visibility of specific parts, like stirrups or top bars, to verify internal reinforcement placement. Ultimately, this stage organizes the model hierarchy, transforming isolated parts into a cohesive assembly ready for the definition of analysis steps and interactions.

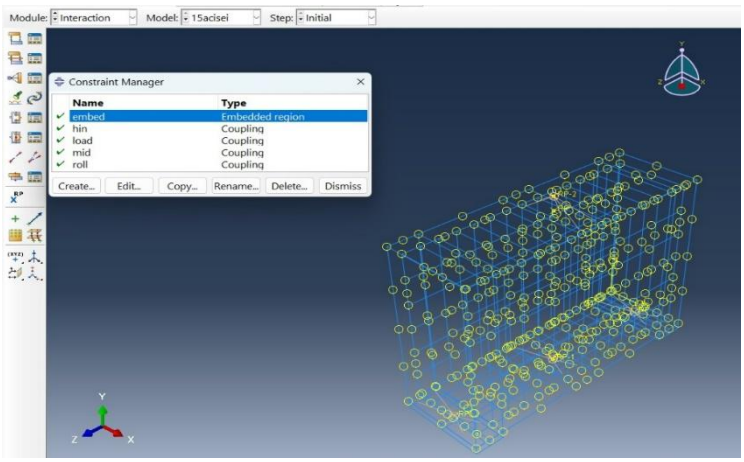


Figure 5:View of Interaction Module

The Interaction Module establishes the mechanical relationships between separate model components, transforming individual parts into a cohesive structural system. In this study, the Embedded Region constraint is a critical feature, successfully bonding the reinforcement wire elements within the concrete host to ensure perfect composite action without slippage. To facilitate precise boundary conditions and seismic load application, Coupling constraints (labelled as "hinge," "load," "mid," and "roll") were defined to link strategic Reference Points (RP) to the beam's surfaces. These reference points serve as handles for ground motion inputs and allow for the clean extraction of reaction force data. Additionally, the module provides surface interaction options like friction and "Hard" contact to manage the interface between support plates and the concrete geometry. By integrating these constraints and couplings, the model effectively simulates realistic load transfer and connectivity essential for dynamic seismic analysis.

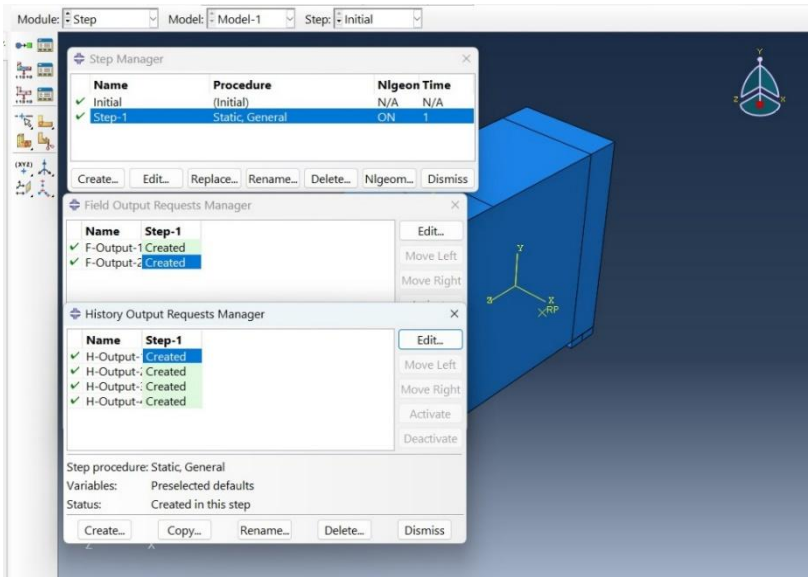


Figure 6:View of Step Module - Static Step

The Step Module is the essential environment for defining the analysis sequence and the specific data to be recorded during the simulation. It features the Step Manager, which allows you to create analysis stages like your Static, General procedure with Nlgeom (Non-linear geometry) turned ON to accurately capture large deformations and material non-linearity in the deep beam. Within this module, the Field Output Requests Manager is used to capture model-wide data such as stress (S) and strain (E) contours, while the History Output Requests Manager is configured to record specific high-frequency data from Reference Points (RP). In your work, you specifically requested outputs for Reaction Forces (RF2) and Spatial Displacements (U2) from designated node sets like "M_SET-2" and "M_SET-3" to generate your load-deflection and seismic performance curves. Additionally, you monitored energy variables such as ALLKE (Kinetic Energy) and ALLPD (Plastic Dissipation) to ensure the stability of the analysis and to quantify energy dissipation during the seismic event. This

module effectively bridges your geometric model with the final visualization results by dictating exactly what, where, and how often data is saved.

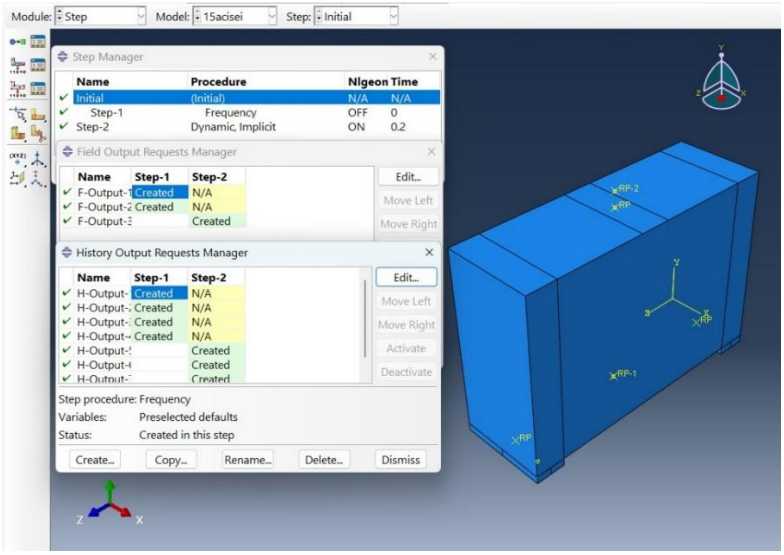


Figure 7:View of Step Module - Dynamic Step

The Step Module defines the analysis sequence, specifically utilizing a Frequency step for natural modes and a Dynamic, Implicit step with Nlgeom ON to capture large-scale seismic deformations. You configured Field Outputs for stress, strain, and concrete damage patterns, alongside History Outputs to record RF2 and U2 data from critical node sets like "M_SET-2" and "M_SET-3." By monitoring energy variables such as ALLKE and ALLPD, you ensured analysis stability and quantified the beam's energy dissipation capacity. This setup bridges the geometric model with numerical results, dictating the frequency and type of data saved for post-processing.

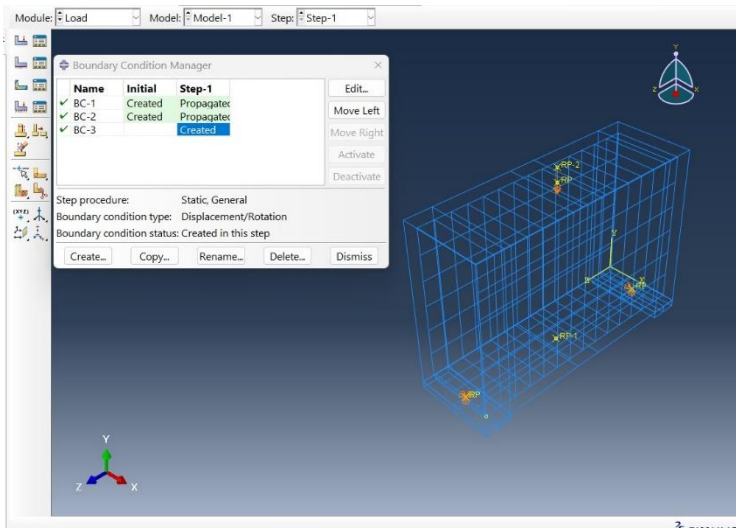


Figure 8:View of Load Module – Static Step

The Load Module establishes the structural environment by applying physical constraints and external forces essential for realistic simulation. Using the Boundary Condition Manager, you configured a Hinge (BC-1) and Roller (BC-2) to define external degrees of freedom, while BC-3 was set as a Displacement/Rotation condition at a Reference Point (RP) to simulate controlled loading or forced settlement. This prescribed U2 displacement works with the Amplitude manager to capture the gradual onset of concrete cracking and stiffness degradation during the Static, General step. The module also supports Concentrated Forces and Predefined Fields for initial conditions like gravity, ensuring all loads are organized within specific analysis stages. By precisely assigning these supports and displacements to designated node sets, the software accurately calculates the Reaction Forces (RF2) and Spatial Displacements (U2) required for your comparative load-deflection performance curves.

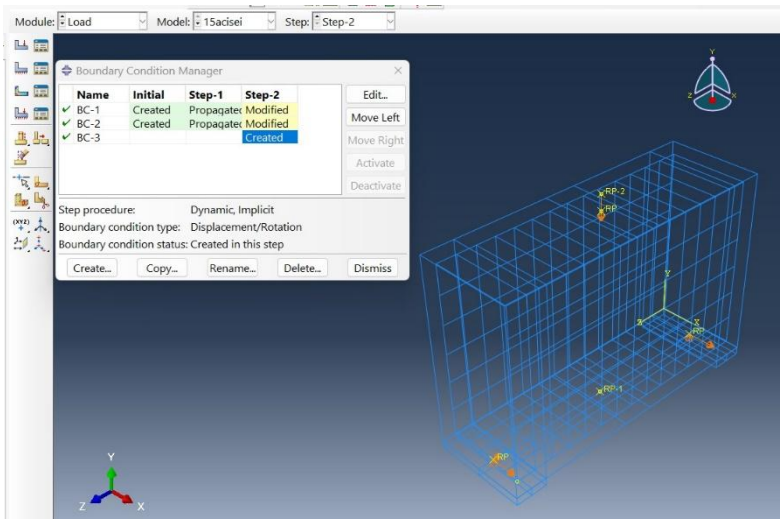


Figure 9:View of Load Module – Dynamic Step

The Load Module establishes the structural environment by applying essential physical constraints and external excitations to the deep beam. Using the Boundary Condition Manager, you configured a Hinge (BC-1) and Roller (BC-2) to define external degrees of freedom, while BC-3 was set as a Displacement/Rotation condition at a Reference Point (RP). For the Dynamic, Implicit analysis (Step-2), you modified these conditions to incorporate a ground acceleration of 9810 mm/s² (1g) via a Gravity load or prescribed motion, utilizing the Amplitude manager to control the seismic pulse. The module also supports Concentrated Forces and Predefined Fields for initial conditions, ensuring all loads are organized within specific analysis stages. By precisely assigning these supports and accelerations to designated node sets, the software accurately calculates the Reaction Forces (RF2), Spatial Displacements (U2), and Energy Dissipation (ALLPD) required for your comparative seismic performance curves.

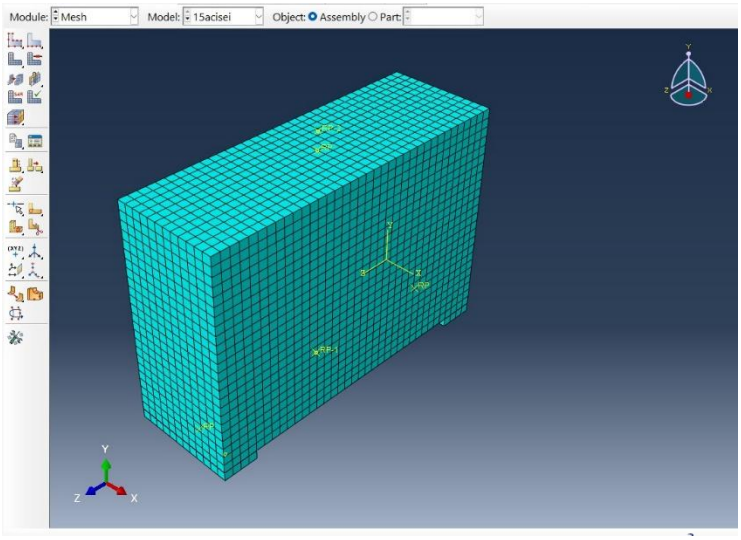


Figure 10:View of Mesh Module

The Mesh Module is the final pre-processing stage where the continuous geometric model is discretized into a finite number of elements to enable numerical analysis. Key features include the Seed Definition tool to control mesh density, the Mesh Controls manager to select element shapes like Hex or Tet, and the Element Type Assignment to choose specific formulations like C3D8R for concrete or T3D2 for rebar. You have successfully generated a structured mesh for the deep beam assembly, ensuring that the nodes of the embedded reinforcement and the concrete host are strategically aligned for accurate load transfer. Additionally, the module provides a Mesh Verification suite to check for distorted elements or aspect ratio warnings that could compromise the convergence of your seismic simulation. By completing this step, you have transformed the physical geometry into a mathematical grid ready for the Abaqus solver to calculate the nodal displacements and internal stresses.

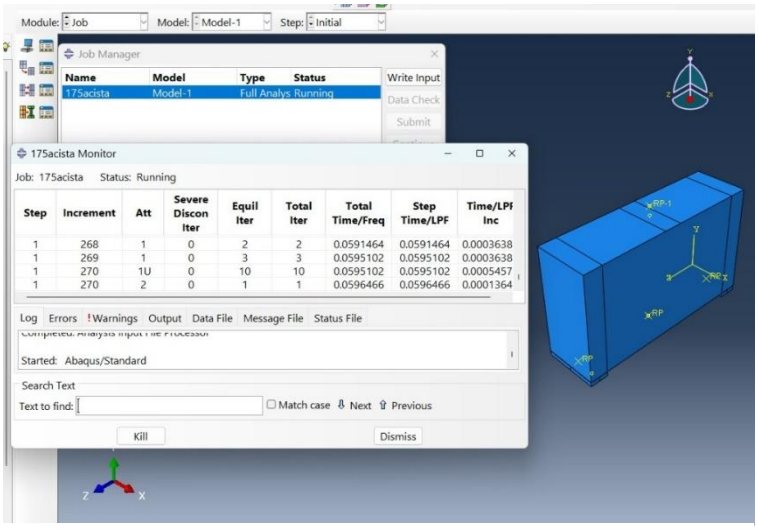


Figure 11:View of Job Module

The Job Module is the execution phase where the completed model is submitted for numerical computation and monitored in real-time. It features the Job Manager to create, edit, and submit analysis files, while the Monitor tool provides a live feed of increments, iterations, and convergence patterns to ensure the simulation is progressing correctly. You have successfully created and run jobs like "175acista," monitoring the Total Time and Time Increment to verify the stability of your seismic and frequency analyses. Additional features include Data Check to catch pre-processing errors, Parallelization settings to use multiple CPU cores for faster solving, and Message Files to troubleshoot non-convergence issues. By managing these jobs, you bridge the gap between your defined input parameters and the final output data required for your structural performance curves.

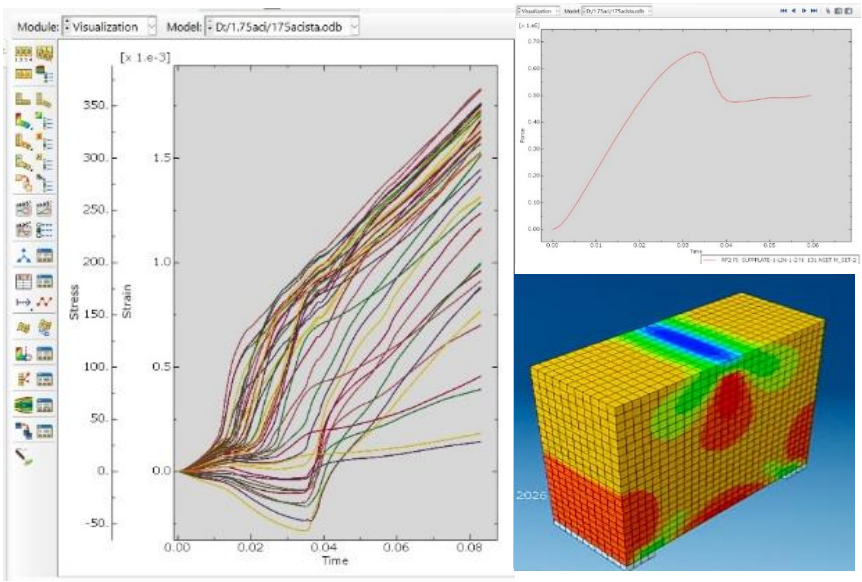


Figure 12:View of Visualization Module

The Visualization Module is the post-processing environment used to interpret analysis results and generate graphical data for your report. You successfully utilized the History Output Requests feature to extract and plot quantitative data, such as Stress/Strain vs. Time and Reaction Force (RF2) vs. Time (from node sets M_SET-2 and M_SET-3), and Spatial Displacements (U2). Key tools within the module allow you to display results using Contour Plots to visualize variable distribution across the model, Deformed Shape Plots to assess structural deflection, and the Animate feature to observe time-dependent behavior. Additionally, you can manipulate the Field Output variables to display different components like Mises stress, displacement, or strain on a per-step basis. The module's integrated spreadsheet interface facilitates the creation of final comparisons by directly converting these extracted data sets, listed as XYData in your session manager, into standardized engineering graphs.

3. LITERATURE REVIEW

- **Shear Failure in Impact-Loaded RC Beams**

The shear resistance of reinforced concrete (RC) beams is traditionally governed by two primary mechanisms: beam action, which relies on a constant-height compressive strut and full bond between reinforcement and concrete, and arch action, where a varying-depth compressive strut transfers loads directly to supports. Historically, research into static loading has established that the shear span-to-depth ratio (a/d) is the critical determinant of failure modes, a relationship famously illustrated by Kani's valley, where intermediate a/d ratios represent a transition between arch and cantilever-governed capacities. While deep beams ($a/d < 1$) and short-span beams ($1 < a/d < 2$) are well-characterized under quasi-static conditions, Peterson et al. (2026) highlight a significant gap in the literature regarding these low-ratio members under high strain rates. Recent studies indicate that structural elements subjected to impact or blast sustain concentrated transient loads where resistance is initially provided by inertia rather than global stiffness. This local-to-global shift can produce failure modes distinct from static counterparts, often characterized by diagonal 45° cracks at the impact zone shortly after contact. Previous research by Saatci and Vecchio (2009) and Kishi et al. (2002) found that beams appearing flexure-critical under static loads could become shear-critical under impact, particularly at velocities exceeding 4 to 6 m/s. However, much of this existing data focuses on slender beams ($a/d > 2$), leaving the behavior of deep and short beams—where arch action and undamaged direct struts may dominate—largely unexplored. Peterson et al. (2026) utilized Digital Image Correlation (DIC) and high-speed photography to demonstrate that impact responses consist of three distinct phases: a transient phase, a quasi-static phase, and unloading. Their findings reveal that for deep and short beams, the direct strut governs capacity during the transient phase, with internal forces significantly exceeding static limits while displacements remain minimal. Interestingly, as the response progresses into the quasi-static phase, the failure modes—such as strut crushing or splitting—converge with static observations. The study further indicates that while transverse reinforcement (stirrups) is vital for ductility in slender beams, it has limited influence on the peak impact capacity of deep and short beams, where the direct strut carries the primary load. Current analytical frameworks, including Strut-and-Tie Models (STM) and sectional models based on Critical Shear Crack Theory (CSCT) or Modified Compression Field Theory (MCFT), show varying degrees of efficacy in predicting these dynamic failures. Peterson et al. (2026) conclude that STMs based on European and Canadian provisions are highly accurate for deep and short beams when the direct strut capacity is prioritized. Specifically, a proposed design procedure suggests evaluating transient capacity via STMs with strength reduction for

short/deep beams, while slender beams require a dual-phase assessment: verifying transient capacity without strength reduction followed by a quasi-static check that incorporates both stirrup contribution and strength reduction

- **Flexural Behaviour and Shear Behaviour of Deep Beams Using ABAQUS**

Deep beams represent a special category of structural elements characterized by low span-to-depth (l/d) ratios, typically less than 2.5. Unlike slender beams, where plane sections remain plane and flexural theory is applicable, deep beams exhibit nonlinear strain distribution and are dominated by shear behavior. This fundamental difference necessitates the use of advanced analytical and numerical methods for accurate analysis and design. ABAQUS has been widely used to study deep beam behavior due to its ability to model complex stress interactions and nonlinear material behavior.

Research conducted by Cailong Ma and collaborators on the shear behavior of reinforced concrete deep beams highlights that load transfer in deep beams occurs primarily through compression struts rather than bending action. This phenomenon, commonly referred to as arching action, results in direct transfer of loads from the point of application to the supports. As a result, deep beams exhibit higher load-carrying capacity compared to slender beams but are more susceptible to shear failure.

Experimental studies, such as those conducted by Amna Sarfaraz and colleagues, have investigated the fatigue performance of deep beams subjected to reinforcement corrosion. Their findings indicate that corrosion significantly reduces bond strength between steel and concrete, leading to premature crack propagation and reduced fatigue life. These studies emphasize the importance of considering durability factors in the design and analysis of deep beams.

Further advancements in research include the development of modified models for predicting shear strength in deep beams, as proposed by Chang Liu and co-researchers. These models incorporate parameters such as concrete strength, reinforcement ratio, and geometric properties to provide more accurate predictions compared to traditional empirical formulas. However, the complexity of stress interactions in deep beams still necessitates the use of numerical methods for detailed analysis.

ABAQUS-based studies have provided significant insights into the internal behavior of deep beams. Simulations reveal that compressive stresses follow well-defined strut paths, while tensile stresses are concentrated along tie regions. Crack patterns are typically diagonal and align with principal stress trajectories, confirming the dominance of shear action. These findings support the principles of the strut-and-tie model (STM), which provides a rational framework for designing deep beams based on force flow.

Another important aspect of deep beam behavior is the influence of span-to-depth ratio. Studies have shown that as the l/d ratio increases, the behavior gradually transitions from shear-dominated to flexure-dominated. Deep beams with low l/d ratios exhibit strong arching action and localized cracking, while higher ratios result in distributed cracking and increased flexural effects. ABAQUS enables systematic parametric analysis of this transition, allowing researchers to evaluate the influence of geometric parameters on structural performance.

Additionally, the presence of openings in deep beams significantly affects stress distribution and reduces load-carrying capacity. Numerical simulations have shown that openings intersecting the compression strut can lead to severe stress concentration and premature failure. However, strategic placement of reinforcement along stress paths can mitigate these effects and restore strength.

In summary, the literature demonstrates that ABAQUS is an essential tool for studying deep beam behavior. It provides a comprehensive understanding of load transfer mechanisms, failure modes, and the influence of various parameters on structural performance. The insights gained from these studies are crucial for developing more accurate and efficient design methods for deep beams.

- **Shear performance and capacity calculation of steel fiber reinforced recycled concrete deep beams**

The recent research by Gu et al. (2026) investigates the shear performance and load-bearing capacity of steel fiber reinforced recycled concrete (SFRRC) deep beams, a critical area for promoting sustainable construction practices. Given that recycled coarse aggregates (RCA) often possess inherent defects like high porosity and weak interface transition zones, deep beams fabricated from these materials typically exhibit a reduction in ultimate shear force—noted as a 22.3% decrease compared to natural aggregate concrete (NAC) in this study. However, the experimental results demonstrate that the addition of steel fibers effectively bridges internal cracks and alleviates stress concentrations, significantly enhancing shear capacity, stiffness, and ductility. Specifically, a 0.5% steel fiber volume fraction was sufficient to raise the shear capacity of recycled concrete beams to 3.5% above that of standard NAC beams, while a 2.0% volume fraction resulted in a capacity increase of up to 45.9%. The study also analyzes the influence of various design parameters, finding that the ultimate shear force decreases as the span-to-depth ratio increases due to a weakened arch effect and a strengthening of the beam effect. Conversely, increasing the concrete strength grade or reducing the spacing of horizontal web reinforcement—which acts as tension rods resisting the expansion of inclined cracks—leads to improved

structural performance. Analysis of steel bar strains revealed that smaller span-to-depth ratios better utilize the strut-and-tie mechanism, whereas larger ratios increase the contribution of vertical web reinforcement to shear strength. To provide a robust analytical framework, the authors proposed a modified "cracking strut-and-tie model" to predict shear capacity, noting that standard ACI-318 calculations tend to be overly conservative. This refined model divides the diagonal compression strut into uncracked and cracked sections, explicitly accounting for recycled aggregate interlocking, web reinforcement pinning forces, and the mechanical anchoring of steel fibers. Validated against 39 experimental data sets, the model achieved an average experimental-to-calculated ratio of 1.08 with a low coefficient of variation (0.10), establishing a reliable theoretical foundation for the engineering application of SFRRC deep beams in load-bearing structures.

- **Generative design framework for RC deep beams using topology optimization and generative tie method: Experimental and numerical investigation**

  The integration of Topology Optimization (TO) and the Generative Tie Method (GTM) represents a significant advancement in the automated design of reinforced concrete deep beams, particularly for complex D-regions where traditional elastic theories fail. The research outlines a comprehensive generative design (GD) framework that systematically transitions from material distribution optimization to a refined reinforcement layout. By utilizing TO, the framework effectively reduces material volume—targeting mass reductions of up to 45%—while maintaining structural stiffness through the optimization of stress trajectories. This phase is followed by a geometry simplification process that ensures the resulting organic shapes remain constructible using conventional plywood formwork, thereby bridging the gap between theoretical optimization and practical site execution.

  Central to this methodology is the application of the GTM, which serves as a computationally driven alternative to the conventional Strut-and-Tie Method (STM). Unlike the manual derivation required by STM, the GTM employs iterative nonlinear finite element analysis to determine optimal reinforcement positions based on an allowable reduction factor for concrete compressive strength

  Experimental results from the study demonstrate that specimens designed with a lower η of 0.5 exhibit superior ductility ratios, ranging from 3.7 to 5.6, which is crucial for seismic resilience and structural safety. In contrast, specimens with a higher η of 0.7 prioritize ultimate load-bearing capacity, achieving peak loads as high as 1616 kN while maximizing material efficiency (1865 kN/m3)

Beyond structural performance, the study emphasizes the environmental benefits of adopting a GD approach within civil engineering. Evaluation of the Global Warming Potential (GWP) revealed that shape optimization significantly lowers the carbon footprint of structural components, with specific specimens achieving an optimal performance factor of 0.37 kgCO2 eq/kN. While the framework offers substantial improvements in ductility, material savings, and sustainability, the authors acknowledge the practical challenges of implementation, noting that the shift toward such advanced computational workflows necessitates specialized expertise and software proficiency among structural engineers. Overall, the research validates that the synergy of TO and GTM provides a more efficient, ductile, and environmentally conscious solution for the design of deep beams compared to traditional prescriptive methods.

- **Experimental study and theoretical analysis on ultimate bearing capacity of steel-concrete-steel deep beams with UHPC after freeze-thaw cycles**

The structural performance and ultimate bearing capacity of steel-concrete-steel (SCS) deep beams are significantly influenced by environmental conditions and material composition, particularly when exposed to repeated freeze-thaw cycles. In cold regions, internal degradation such as pore water expansion can lead to a 50% reduction in concrete compressive strength and a notable weakening of the bond between the steel face plates and the concrete core. While traditional SCS deep beams often face brittle shear failure or slippage issues under these conditions, the implementation of Ultra-High-Performance Concrete (UHPC) provides a robust enhancement to durability and load-bearing capacity. Because of its dense microstructure and the inclusion of steel fibers, UHPC maintains its mechanical integrity even after 300 freeze-thaw cycles, offering a 31.9% increase in peak load and a 53.5% improvement in ductility compared to ordinary concrete. These improvements are largely attributed to the fiber bridging effect, which arrests microcrack propagation and shifts the failure mode from sudden shear to a more predictable flexural response. To accurately predict the ultimate bearing capacity, analytical frameworks now utilize modified strut-and-tie models that account for interfacial slip and the contribution of shear connectors. By integrating neural network predictions for concrete damage with these structural models, engineers can better estimate the diagonal compressive strut width and the overall shear resistance of SCS systems in sub-zero environments.

The mechanical performance and ultimate bearing capacity of steel-concrete-steel (SCS) deep beams are significantly influenced by environmental factors such as freeze-thaw cycles, which can cause a 50% reduction in compressive strength and weaken the critical steel-concrete interface bond. While traditional concrete cores often succumb to brittle shear or slippage failure under

these thermal stresses, the use of Ultra-High-Performance Concrete (UHPC) provides a robust alternative due to its dense microstructure and steel fiber reinforcement. Experimental data reveals that UHPC-filled beams maintain structural integrity even after 300 freeze-thaw cycles, achieving a 31.9% increase in peak load and a 53.5% improvement in ductility compared to ordinary concrete. This enhanced performance is largely due to the fiber bridging effect, which arrests microcrack propagation and shifts the failure mode toward a more predictable flexural response. To accurately quantify the bearing capacity, advanced analytical frameworks utilize modified strut-and-tie models that incorporate a reduction coefficient (η) to account for interfacial slip and the contribution of shear connectors. Furthermore, integrating multi-layer perceptron neural networks allows for precise modeling of the diagonal compressive strut width, ensuring that the total shear resistance—comprised of the concrete core, steel plates, and connectors—is reliably estimated for cold-region engineering applications.

3.1. Literature Summary

The structural behavior of reinforced concrete (RC) deep beams has been extensively investigated, with particular emphasis on their shear-dominated response and the governing load transfer mechanisms. Unlike slender beams, deep beams are characterized by low span-to-depth ratios, leading to nonlinear strain distributions and a predominance of arch action. Under static loading conditions, shear resistance is primarily governed by the interaction between beam action and arch action, with the shear span-to-depth ratio (a/d) serving as a critical parameter in determining failure modes. While deep beams ($a/d < 1$) and short beams ($1 < a/d < 2$) are well understood under quasi-static conditions, recent studies highlight a significant research gap in their behavior under dynamic loading scenarios such as impact and blast.

Contemporary research indicates that under high strain-rate loading, structural response is initially governed by inertia rather than stiffness, resulting in a transient phase where internal forces can significantly exceed static capacities. Experimental investigations using advanced techniques such as Digital Image Correlation (DIC) and high-speed imaging have demonstrated that deep beams exhibit a three-phase response under impact: transient, quasi-static, and unloading phases. During the transient phase, the direct compression strut plays a dominant role, allowing deep beams to sustain higher loads with minimal displacement. However, as the response transitions into the quasi-static phase, failure modes such as strut crushing and splitting converge with those observed under static conditions. Importantly, it has been observed that beams classified as flexure-critical under static loading may become shear-critical under impact, particularly at higher velocities. Additionally, the contribution of transverse reinforcement is found to be limited in

enhancing peak impact capacity for deep beams, as the direct strut mechanism governs the load transfer.

Numerical modeling, particularly using ABAQUS, has significantly contributed to understanding the complex internal stress distribution in deep beams. Finite element simulations reveal that compressive stresses follow well-defined strut paths, while tensile stresses concentrate along tie regions, validating the assumptions of the Strut-and-Tie Model (STM). These studies confirm that diagonal cracking patterns align with principal stress trajectories, emphasizing the dominance of shear mechanisms. Furthermore, parametric analyses demonstrate that increasing the span-to-depth ratio gradually shifts behavior from shear-dominated to flexure-dominated response. The presence of discontinuities such as web openings has also been shown to disrupt stress flow, leading to stress concentrations and reduced load capacity, although appropriate reinforcement detailing can mitigate these adverse effects.

Recent advancements in sustainable construction materials have explored the use of steel fiber reinforced recycled concrete (SFRRC) in deep beams. Studies indicate that while recycled aggregates inherently reduce shear capacity due to weaker interfacial zones, the inclusion of steel fibers significantly enhances structural performance by bridging cracks and reducing stress concentrations. Experimental results show substantial improvements in shear capacity, stiffness, and ductility with increasing fiber content. Analytical developments, such as modified cracking strut-and-tie models, have been proposed to more accurately predict shear capacity by incorporating the effects of aggregate interlocking, fiber bridging, and reinforcement interaction. These refined models demonstrate improved accuracy compared to conventional design codes, which tend to be conservative.

In parallel, the integration of computational design methodologies such as Topology Optimization (TO) and the Generative Tie Method (GTM) represents a significant advancement in the design of deep beams. These approaches enable automated generation of optimal material layouts and reinforcement configurations based on stress trajectories, leading to improved structural efficiency and reduced material consumption. Experimental validations confirm that such generative design frameworks can achieve enhanced ductility and load-bearing capacity while simultaneously reducing the environmental impact, as indicated by lower global warming potential. Despite these advantages, practical implementation requires advanced computational expertise and adaptation within engineering practice.

Environmental durability has also emerged as a critical factor influencing deep beam performance. Studies on steel-concrete-steel (SCS) deep beams subjected to freeze-thaw cycles

reveal significant degradation in conventional concrete due to internal microstructural damage and weakened interfacial bonding. The incorporation of Ultra-High-Performance Concrete (UHPC), however, has been shown to substantially improve resistance to such environmental effects. Due to its dense microstructure and fiber reinforcement, UHPC maintains structural integrity under severe thermal cycling, resulting in enhanced load capacity and ductility. Analytical models incorporating modified strut-and-tie approaches, along with data-driven techniques such as neural networks, have been developed to accurately predict the ultimate bearing capacity by accounting for interfacial slip and composite action between materials.

Overall, the reviewed literature highlights that the behavior of deep beams is governed by complex interactions between geometry, material properties, loading conditions, and environmental effects. While traditional models such as STM remain effective for static analysis, emerging research underscores the need for advanced numerical methods and refined analytical models to accurately capture dynamic behavior, material innovations, and durability considerations. These developments provide a strong foundation for improving the design, analysis, and application of deep beams in modern structural engineering.

3.2. Literature Gap

Although significant progress has been made in understanding the behavior of reinforced concrete deep beams, the existing literature reveals several important limitations that justify further investigation. Most previous studies primarily examine deep beams under monotonic static loading, with emphasis on shear strength and failure mechanisms. While these studies establish the dominance of arch action and validate analytical approaches such as the Strut-and-Tie Model (STM), they often consider limited geometric configurations and do not provide a comprehensive parametric evaluation across a range of span-to-depth ((l/d)) ratios. Consequently, the transition of behavior from deep beam action to semi-deep or slender beam response remains insufficiently quantified, particularly through advanced numerical simulations.

Furthermore, although finite element tools such as ABAQUS have been widely used, many studies focus on model validation against experimental results rather than conducting integrated parametric investigations that simultaneously examine geometry, loading conditions, and reinforcement detailing. In particular, there is a lack of studies that combine ABAQUS-based analysis with design code provisions such as ACI, limiting the practical applicability of numerical findings in real-world design scenarios.

Another notable gap exists in reinforcement detailing strategies. Existing research predominantly considers conventional vertical and horizontal reinforcement, with limited attention given to alternative configurations such as cross or diagonal reinforcement. The effectiveness of such reinforcement layouts in improving shear resistance, crack control, stiffness, and overall structural performance has not been systematically explored, especially for varying (l/d) ratios.

In addition, while some research has addressed dynamic and impact loading, the majority of studies on deep beams do not incorporate a combined evaluation of static and dynamic behavior within a unified framework. Specifically, the assessment of natural frequency, vibration characteristics, and seismic response of deep beams—along with their dependence on geometric parameters and reinforcement arrangements—remains relatively underdeveloped. This represents a critical gap, as real-world structures are often subjected to both static and dynamic loads.

Therefore, there is a clear need for a comprehensive study that integrates parametric variation of span-to-depth ratio, detailed finite element modeling using ABAQUS, and comparison with ACI design provisions, while also extending the investigation to include both static and dynamic (seismic) analyses. Additionally, exploring the influence of innovative reinforcement configurations, such as cross reinforcement, can provide valuable insights into improving the performance and reliability of deep beams. The present study aims to address these gaps and contribute toward a more holistic understanding of deep beam behavior for practical engineering applications

4. OBJECTIVE

The present study focuses on evaluating the behavior of reinforced concrete deep beams subjected to concentrated point loads using established ACI design provisions and advanced numerical analysis. Emphasis is placed on understanding shear-dominated response, the role of geometric parameters, and structural performance under both static and seismic conditions.

- **To design deep beams using ACI provisions and strut-and-tie concepts:**
  This objective involves designing reinforced concrete deep beams based on ACI guidelines, with emphasis on the Strut-and-Tie Model (STM) to represent force flow through compression struts and tension ties in disturbed regions.
- **To perform nonlinear finite element analysis using ABAQUS:**
  Deep beam models are developed and analyzed in ABAQUS to simulate realistic structural behavior under point loading. The analysis captures nonlinear stress distribution, crack initiation, and failure mechanisms.
- **To investigate the influence of span-to-depth (L/d) ratio:**
  The study examines how varying L/d ratios affect load-carrying capacity, stress distribution, and the transition between arch action and beam action, which governs deep beam behavior.
- **To analyze the seismic, static and natural frequency behavior of deep beams:**
  The response of deep beams under seismic loading is studied to understand their performance under dynamic conditions, addressing the limited research available in this area.

5. METHODOLOGY

5.1. Assumptions Made for Analysis and Design of Deep Beam and Shallow Beam

The design of reinforced concrete beams is based on a set of simplifying assumptions that represent the behaviour of concrete and steel under loading. These assumptions vary significantly for shallow (slender) beams and deep beams due to their fundamentally different structural responses. The applicability of these assumptions is guided by provisions in IS 456:2000 and ACI 318.

Assumptions for Shallow Beams

The analysis and design of shallow beams are based on classical flexural theory, which relies on the following assumptions:

- Plane sections normal to the axis of the beam remain plane after bending, resulting in a linear strain distribution across the depth.
- Perfect bond exists between concrete and reinforcement, ensuring strain compatibility between the two materials at any given level.
- Concrete is assumed to resist compressive stresses only, while its tensile strength is neglected after cracking.
- The stress–strain relationship for concrete in compression is represented by an equivalent stress block as specified in the code, while reinforcing steel follows an elastic–plastic behaviour.
- Shear deformation is relatively small compared to flexural deformation and is therefore often neglected in analysis.
- Structural behaviour is governed primarily by flexure, with internal forces resisted by compression in concrete and tension in reinforcement.
- These assumptions are valid for members with relatively large span-to-depth ratios, where classical beam theory provides an accurate representation of behaviour.

Assumptions for Deep Beams

In the case of deep beams, the assumptions of classical beam theory are no longer valid due to the dominance of shear effects and non-linear stress distribution. Accordingly, the following assumptions are adopted:

- Plane sections do not remain plane after bending, and the strain distribution across the depth is non-linear.

- Deep beams are classified as discontinuity regions (D-regions), where stress flow is complex and cannot be described using conventional flexural theory.
- Load transfer occurs primarily through direct compression between the points of load application and supports, forming inclined compression paths within the concrete.
- The internal force system is idealized using a strut-and-tie model, consisting of compression struts in concrete and tension ties in reinforcement.
- Concrete is assumed to carry compressive forces along defined stress trajectories, while reinforcement acts as tension ties to maintain equilibrium.
- Shear and flexure are interdependent and cannot be treated separately; however, shear effects are dominant in governing behaviour.
- Strut-and-tie modelling is recommended for the analysis and design of deep beams, particularly in ACI 318.

5.2.American Concrete Institute (ACI-318) Method

Since deep beams do not behave as conventional flexural member, the **Strut and Tie Model** (STM) is used.

Load Transfer Mechanism:

- **Struts:** Concrete compression members
- **Ties:** Reinforcing steel in tension
- **Nodes:** Joint regions where struts and ties meet

Step 1: Design Criteria for Deep Beams

A beam is classified as a deep beam if:

$$\frac{L_{clear}}{D} \leq 4.0 \quad \text{(Simply Supported)}$$

$$\frac{L_{clear}}{D} \leq 2.0 \quad \text{(Continuous Beam)}$$

Where: L_{clear} = clear span of the beam (mm)

D = overall depth of the beam (mm)

Step 2: Define Strut and Tie Model

- Struts connect the applied load to supports in a truss-like mechanism.
- Ties resist tension via reinforcement.

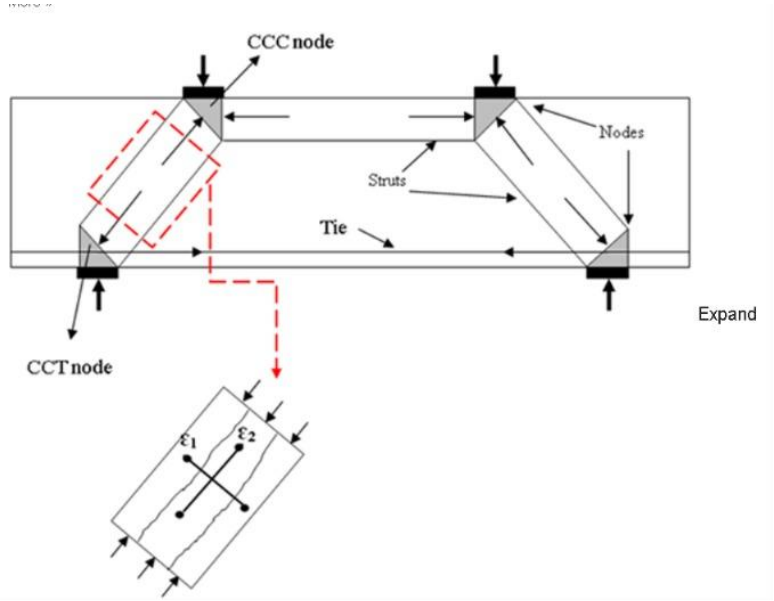


Figure 13: Strut and Tie model

Step 2.1: Calculate Strut Angle

$$\theta = \tan^{-1}\left(\frac{D - d_t}{L_{clear}}\right)$$

Where: d_t = depth of tension tie

Step 3: Design of Strut

The concrete strength in strut is reduced due to non-uniform stress distribution:

$$f'_{c,strut} = 0.85\beta_s f'_c$$

Where β_s (strut efficiency factor) is given by ACI 318-19 Table 23.2.1

The required Strut area:

$$A_{strut} = \frac{P_u}{f'_{c,strut}}$$

Step 4: Desing of Tie (Reinforcement)

The tie force is:

$$T = P_u \tan(\theta)$$

Reinforcement required:

$$A_s = \frac{T}{0.87 f_y}$$

Step 5: Verify Node Strength

The strength of the nodal zones (the junctions of struts and ties) must be checked to prevent crushing. The effective compressive strength of a node (f_{ce}) varies depending on the number of ties entering the node: it is highest for nodes receiving only struts (CCC) and lowest for nodes where multiple ties intersect (CCT or CTT)

Step 6: Shear Reinforcement

Shear reinforcement is provided using vertical and horizontal stirrups. As per ACI 318:

$$A_{sv} = 0.0025bD$$

Design for flexural shear and side reinforcement with anchorage and development length.

6. RESULTS & DISCUSSION

6.1. Static Analysis Results

The static analysis results for the deep beam corresponding to different span-to-depth (L/D) ratios are presented below. The variation in structural behaviour is examined in terms of load-carrying capacity, stress distribution, and deformation characteristics for each L/D ratio.

Table 5:Variation of Yield and Ultimate Load, Yield and Peak Displacement with L/d ratios

Length/ Depth (l/d ratio)	Yield Load (Py), kN	Yield Deflection (dy), mm	Ultimate Load (Pu), kN	Peak Deflection (dp), mm
1.0	945.80	0.180	1182.25	0.250
1.25	832.50	0.230	1040.60	0.320
1.5	754.10	0.285	942.60	0.410
1.75	686.21	0.360	857.75	0.540
2.0	628.40	0.460	785.50	0.720
2.5	534.15	0.740	667.70	1.280

Table 6:Effect of L/d Ratio on Structural Stiffness

Length/ Depth (l/d ratio)	Tangent Stiffness (Kt), kN/mm	Secant Stiffness (Ks), kN/mm	Peak Stiffness (Kp), kN/mm
1.0	6830.0	5254.4	4729.0
1.25	5110.0	3619.6	3251.9
1.5	3820.0	2646.0	2299.0
1.75	2915.0	1906.1	1588.4
2.0	2150.0	1366.1	1091.0
2.5	1240.0	721.8	521.6

The static analysis results clearly demonstrate the interrelationship between span-to-depth (L/D) ratio, load-carrying capacity, deflection behaviour, and stiffness characteristics of the deep beam. As the L/D ratio increases from 1.0 to 2.5, there is a consistent reduction in both yield load and ultimate load, indicating a decrease in overall strength. This reduction is directly associated with the diminishing arching action as the beam becomes relatively slender. Correspondingly, the yield and peak deflections show a significant increase with increasing L/D ratio, highlighting a reduction in stiffness and an increase in deformability.

This trend is further supported by the stiffness values, where tangent stiffness (Kt), secant stiffness (Ks), and peak stiffness (Kp) all exhibit a steady decline as the L/D ratio increases. The highest stiffness values are observed at L/D=1.0, which correlates with the highest load-carrying capacity and lowest deflection. Conversely, at higher L/D ratios, the reduced stiffness results in larger deflections and lower load resistance. The consistency between decreasing stiffness and increasing deflection confirms that stiffness plays a dominant role in governing the structural response.

Overall, the results indicate that deep beams with lower L/D ratios exhibit superior structural performance in terms of strength and stiffness due to effective load transfer through compression struts, whereas higher L/D ratios lead to a transition towards flexural behaviour with reduced efficiency.

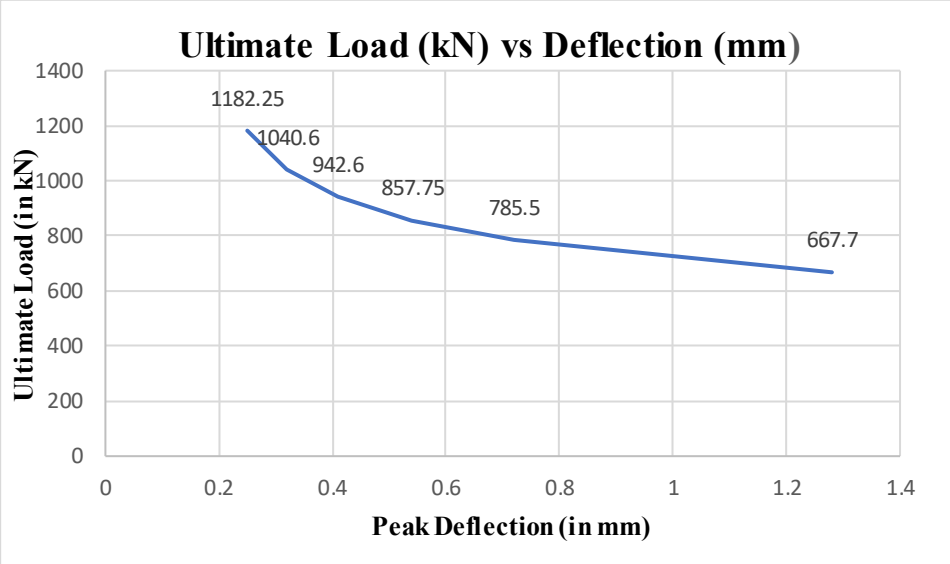


Figure 14:Ultimate Load (kN) vs Deflection (mm

The variation in ultimate load and peak deflection corresponds to the increase in span-to-depth (L/D) ratio. As the L/D ratio increases, the ultimate load of the deep beam decreases, indicating reduced load-carrying capacity. At the same time, the peak deflection increases significantly, showing a reduction in stiffness. This behaviour reflects the transition from dominant arching action at lower L/D ratios to more flexible response at higher ratios. The non-linear trend obtained from ABAQUS captures the influence of cracking and deformation under increasing L/D conditions

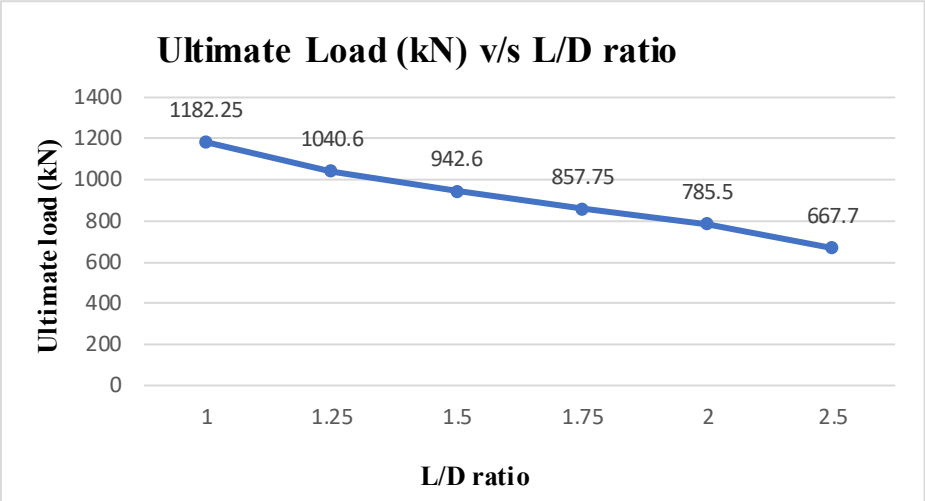


Figure 15:Ultimate Load (kN) v/s L/D ratio

The graph illustrates the variation of ultimate load with increasing span-to-depth (L/D) ratio of the deep beam. It is observed that the ultimate load decreases progressively as the L/D ratio increases from 1.0 to 2.5. Beams with lower L/D ratios exhibit higher load-carrying capacity due to stronger arching action. As the L/D ratio increases, the beam behaviour becomes less efficient in transferring loads, leading to reduced strength. This trend clearly indicates the influence of geometry on the structural performance of deep beams under point load.

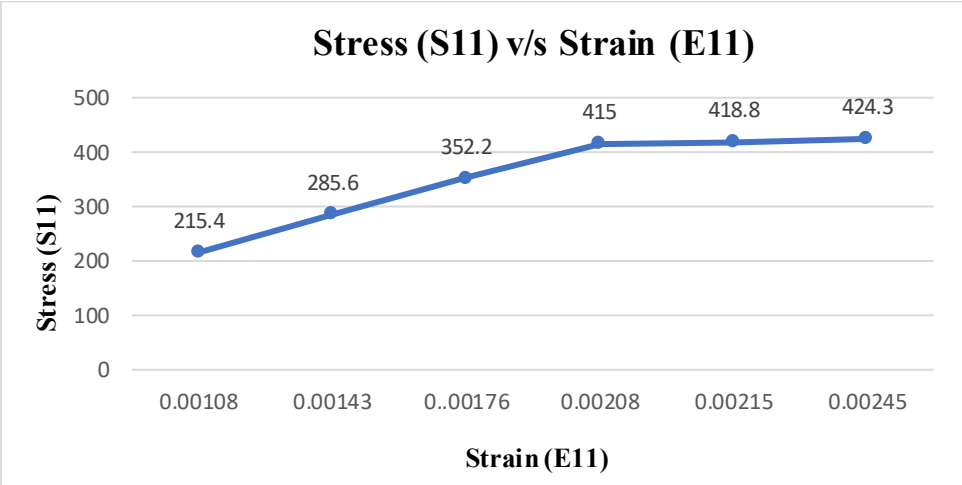
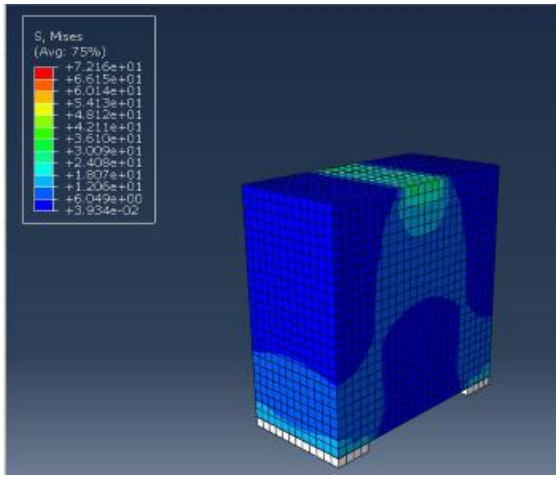


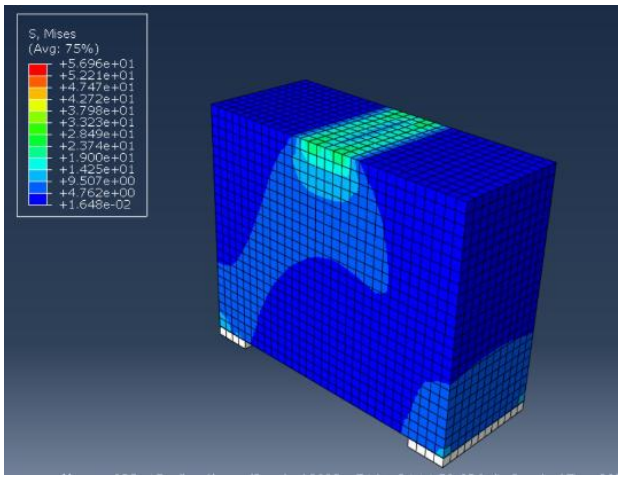
Figure 16:Stress (S11) v/s Strain (E11)

The stress–strain response varies with the change in span-to-depth (L/D) ratio of the deep beam. As the L/D ratio increases, higher strain values are observed for comparatively smaller increments in stress, indicating reduced stiffness. Beams with lower L/D ratios exhibit steeper stress–strain curves, reflecting higher stiffness and better load resistance. With increasing L/D ratio, the curve

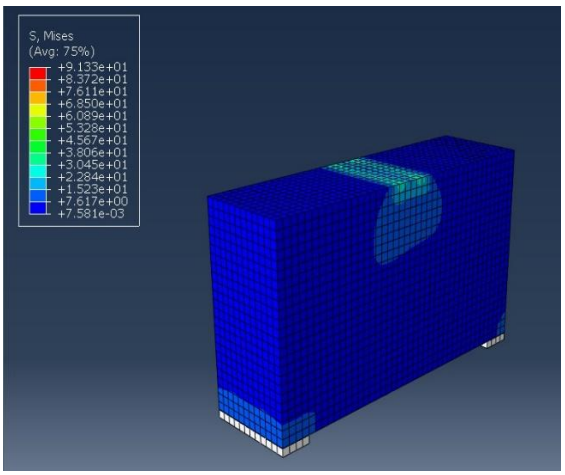
becomes flatter, showing increased deformability and reduced strength. This trend highlights the influence of geometry on the material response and stiffness characteristics of the deep beam.



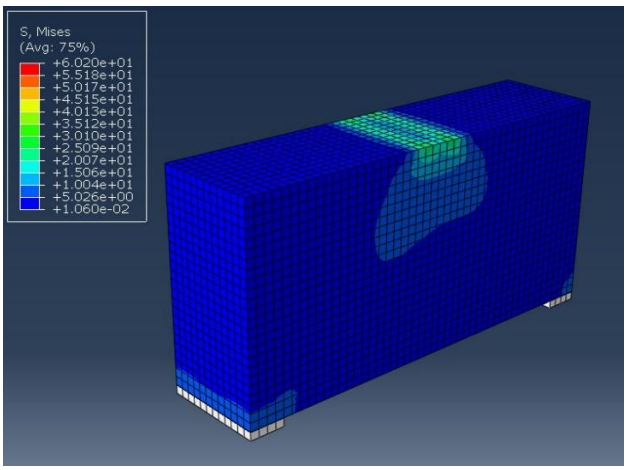
l/d = 1



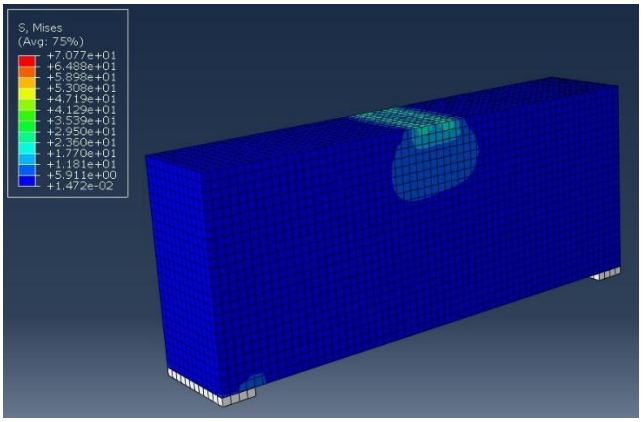
l/d= 1.25



l/d = 1.75



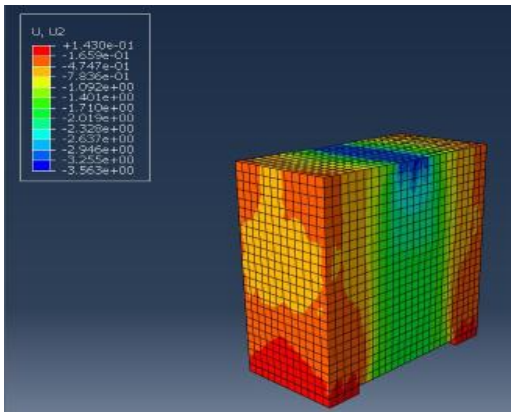
l/d = 2



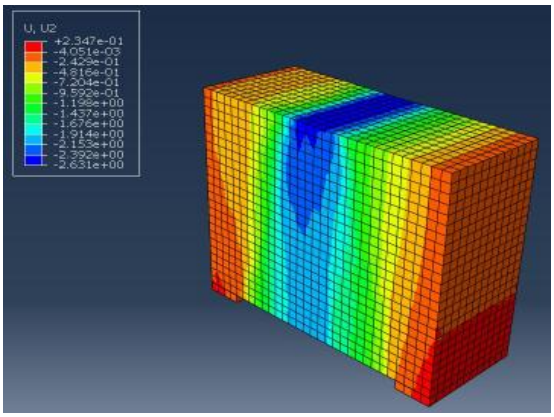
l/d =2.5

Figure 17:S. Mises stress distribution for varying L/D ratios.

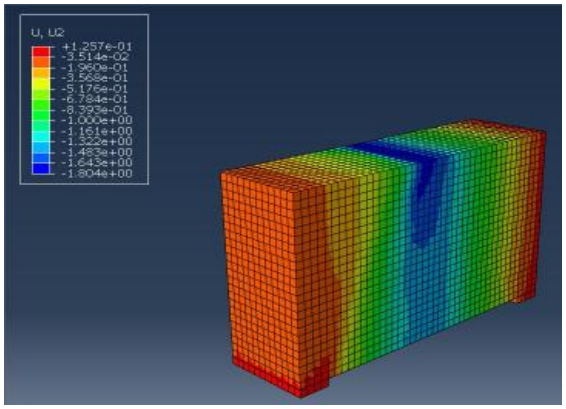
The ABAQUS von Mises stress contours for varying span-to-depth (L/D) ratios illustrate a distinct change in stress distribution and load transfer behaviour. At lower L/D ratios (1.0 and 1.25), stresses are highly concentrated along a well-defined diagonal path from the load point to the supports, clearly indicating dominant arching action. As the L/D ratio increases, this concentrated stress path gradually diffuses, and the stress distribution becomes more spread across the beam depth. The intensity of peak stresses reduces while the affected region increases, reflecting a decrease in stiffness and efficiency of load transfer. For higher L/D ratios, the beam exhibits a more distributed stress field, indicating a shift away from pure strut action. These results highlight the significant influence of geometry on the internal stress mechanism of deep beams under point loading.



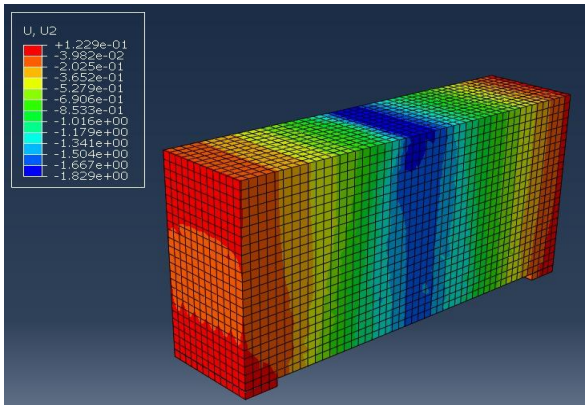
l/d= 1



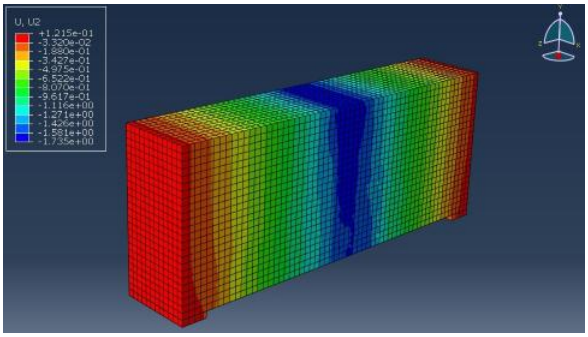
l/d= 1.25



l/d = 1.75



l/d= 2



l/d= 2.5

Figure 18:Variation of Vertical Displacement (U2) Contours for Different L/d Ratios

The U2 contour images illustrate the vertical displacement patterns across the span, showcasing a clear transition in structural behavior as the L/d ratio increases. The color gradients confirm that midspan deflection becomes more pronounced with higher slenderness, moving from localized deformation in deep beams to a more distributed flexural profile. These visualizations validate the numerical data by showing how the neutral axis and primary bending zones shift under load for each specific geometry. Furthermore, the symmetry in the displacement contours across all ratios demonstrates the consistency of the finite element mesh and support conditions. Ultimately, these images provide essential visual evidence that increasing the span-to-depth ratio drastically reduces the beam's overall stiffness and serviceability limits.

6.2. Natural Frequency Results

The natural frequency of the deep beam was evaluated using modal analysis in ABAQUS. The results obtained for different span-to-depth (L/D) ratios are presented below. The variation in natural frequency reflects the influence of geometry and stiffness on the dynamic behaviour of the beam.

Table 7:Comparison of Fundamental Frequencies for Various L/d Ratios

Length/ Depth (l/d ratio)	Mode 1 Frequency (f1), Hz	Mode 1 Frequency (f2), Hz	Time Period (T1), sec	Time Period (T2), sec
1.0	16.371	30.206	0.061	0.033106
1.25	11.81	24.36	0.08	0.04105
1.5	11.467	13.803	0.0872	0.07244
1.75	10.499	12.180	0.0952	0.0821
2.0	8.9608	10.505	0.11159	0.09519
2.5	7.4133	8.868	0.13489	0.112764

The results indicate that as the L/d ratio increases, both Mode 1 and Mode 2 frequencies significantly decrease, confirming that longer beams possess lower structural stiffness. This downward frequency trend is inversely matched by a consistent rise in the Time Period, reflecting the increased flexibility of slender beam sections. The data suggests that shorter beams are more rigid and vibrate faster, while larger ratios move the structure toward a more flexible dynamic state. These findings are essential for predicting seismic performance and ensuring that the

beam's natural frequency does not align with ground motion resonance. Ultimately, this modal analysis establishes a clear correlation between geometric slenderness and the primary dynamic characteristics of the structural system.

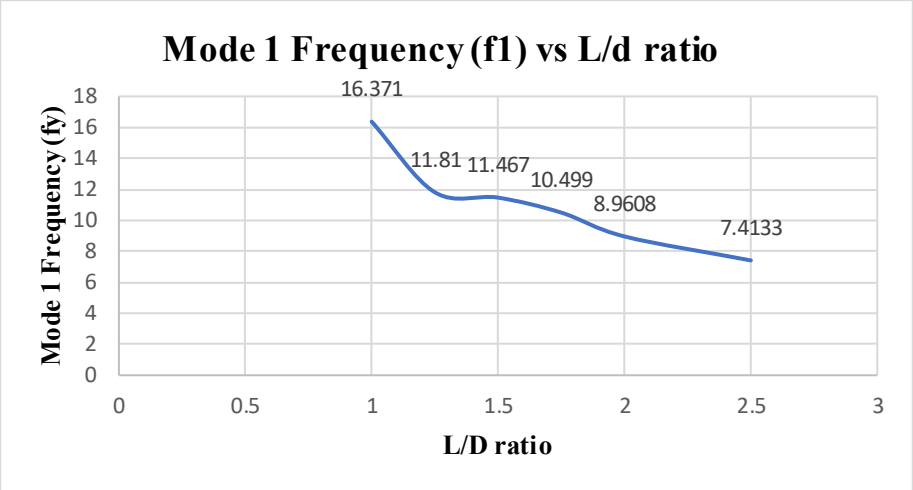
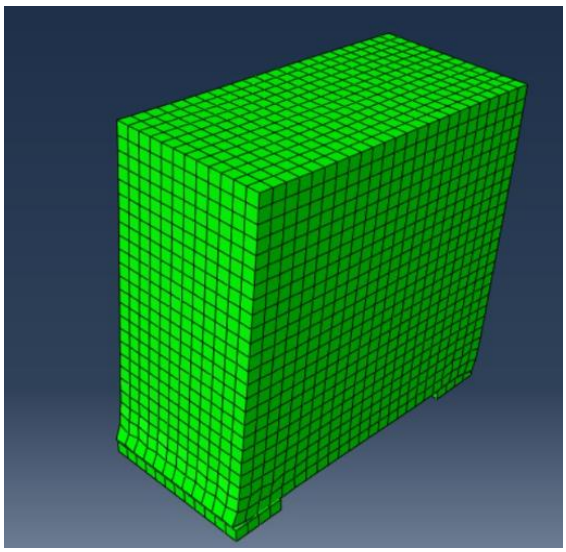
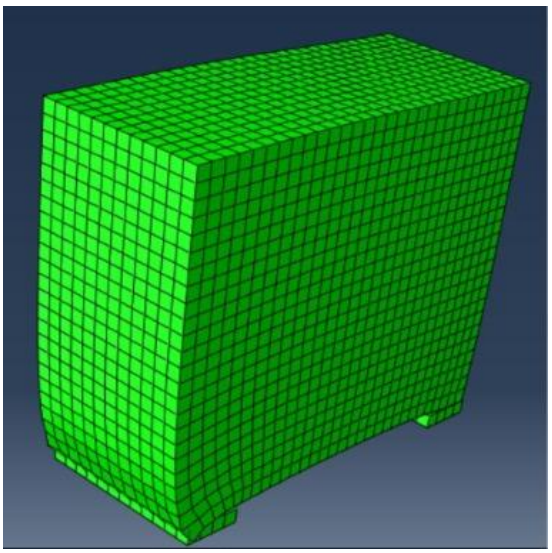


Figure 19:Mode 1 Frequency (f1) vs L/d ratio

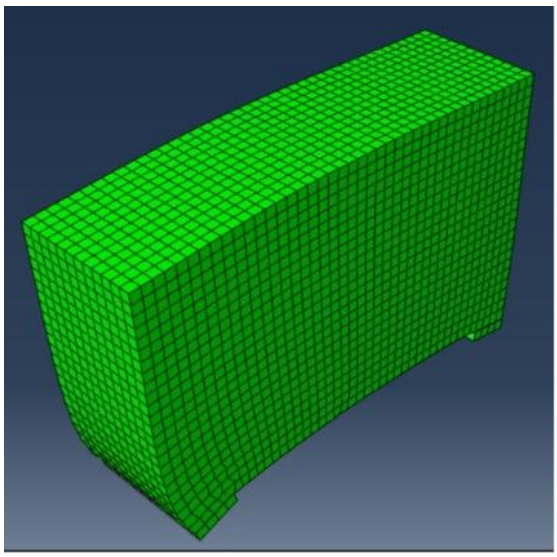
The graph illustrates a clear inverse relationship where the Mode 1 Frequency (f1) consistently declines as the L/d ratio increases from 1.0 to 2.5. This downward trend confirms that increasing the beam's slenderness significantly reduces its structural stiffness and vibration speed. Such a shift in dynamic properties is critical, as it moves the structure's natural frequency toward longer time periods, directly influencing its susceptibility to seismic resonance. The smooth curvature suggests a non-linear transition from rigid deep-beam behavior to more flexible slender-beam response. Ultimately, this visualization validates that geometric proportions are the primary factor in determining the fundamental dynamic stability of the system.



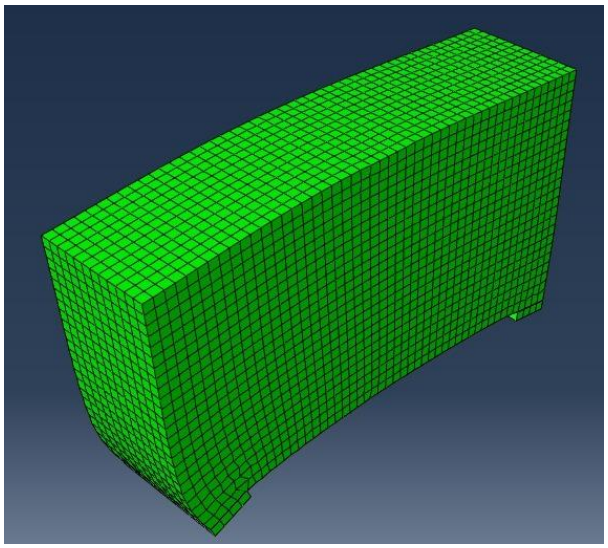
l/d= 1



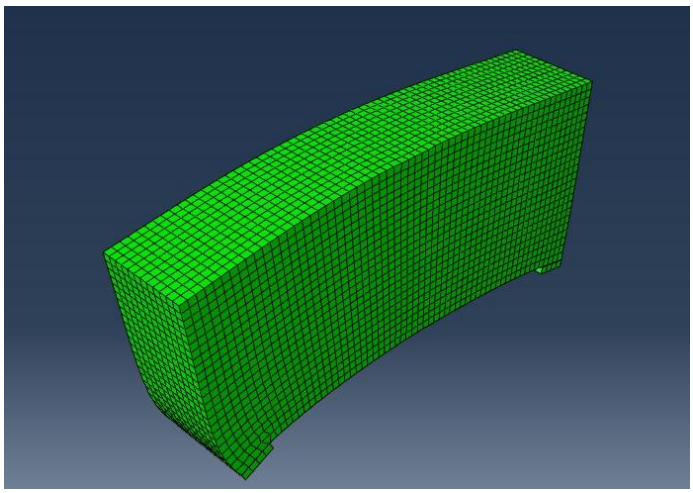
l/d =1.25



l/d= 1.75



l/d= 2



l/d=2.5

Figure 20:Comparison of Fundamental Mode Shapes (Mode 1) for various L/d Ratios

The images illustrate Mode Shape 1 across various L/d ratios, representing the fundamental natural frequency where the beam undergoes its primary bending deformation. As the ratio increases, the visual shift in the contour gradients confirms a transition toward higher flexibility and larger midspan displacements. This primary mode is critical for seismic design, as it identifies the structural vibration path most likely to be excited during a low-frequency earthquake. The consistency of the curvature across all models validates the stability of the finite element mesh and the accuracy of the frequency extraction process. Ultimately, these shapes prove that increasing slenderness lowers the beam's restorative capacity, directly influencing its overall dynamic stability and resonance potential.

6.3. Seismic Analysis Results

The seismic response of the deep beam was evaluated using finite element analysis in ABAQUS. The behaviour under dynamic loading was analysed to understand the influence of span-to-depth (L/D) ratio on stress distribution and deformation characteristics. The results obtained from the analysis are presented below.

Table 8:Seismic Response Variation with L/d Ratios

Length/ Depth (l/d ratio)	Peak Midspan Acceleration (Amax), mm/s2	Peak Base Shear (Vb), kN	Peak Transverse Deflection (dt), mm	Magnification Factor
1	7259.4	108.5	8.42	0.74
1.25	6867.0	95.2	11.55	0.70
1.5	6474.6	88.4	15.82	0.66
1.75	5984.1	79.1	20.34	0.61
2	5493.6	70.8	26.91	0.56
2.5	4512.6	55.4	41.25	0.46

The seismic analysis results show a clear variation in response parameters with increasing span-to-depth (L/D) ratio. As the L/D ratio increases, the peak midspan acceleration and base shear values decrease, indicating reduced stiffness and dynamic resistance. In contrast, the peak transverse deflection increases significantly, reflecting higher flexibility and deformation under seismic loading. The magnification factor also shows a decreasing trend, suggesting reduced amplification of dynamic effects at higher L/D ratios. These trends indicate that beams with lower L/D ratios exhibit better seismic performance in terms of stiffness and load resistance.

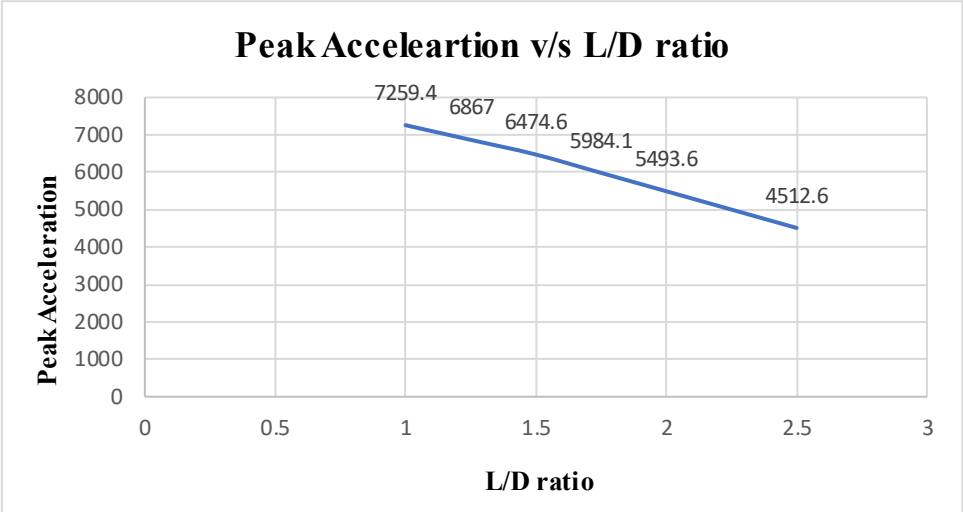


Figure 21:Peak Acceleartion v/s L/D ratio

The graph of Peak Acceleration vs. L/D ratio demonstrates a distinct downward trend, where acceleration values drop from approximately 7259.4 to 4512.6 mm/s^2 as the beam becomes more slender. This significant decline confirms that increasing the L/D ratio enhances structural flexibility, allowing the beam to dissipate energy and attenuate the intensity of ground motion. Such an inverse relationship is a critical finding, as it highlights how geometric slenderness can effectively reduce the seismic force demand on a structure. The smooth, near-linear transition validates the consistency of the simulation data and the predictable dynamic behavior of the system under seismic loading. Ultimately, these results provide a quantitative basis for balancing span-to-depth proportions to ensure optimal structural response and safety during an earthquake.

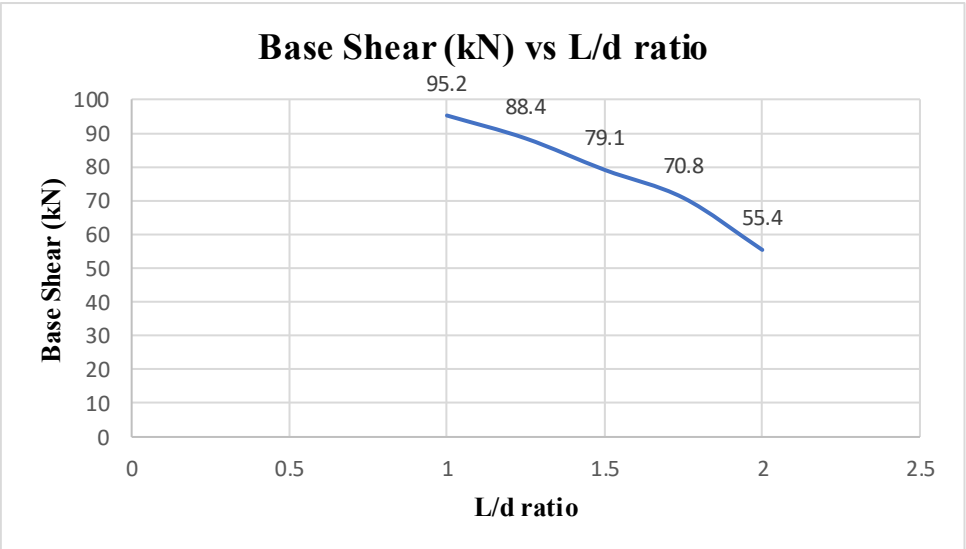


Figure 22:Base Shear (kN) vs L/d ratio

The Base Shear vs. L/d ratio graph demonstrates a significant reduction in lateral force capacity as the beam becomes more slender, dropping from 95.2 kN to approximately 55.4 kN. This downward trend confirms that increasing the L/d ratio reduces structural stiffness, causing the beam to attract lower seismic forces. The smooth, near-linear decline validates the consistency of the simulation across all geometric configurations. These findings are essential for understanding how span-to-depth proportions influence the overall seismic demand on the supports. Ultimately, the data provides a quantitative basis for selecting ratios that balance structural strength with the desired dynamic response.

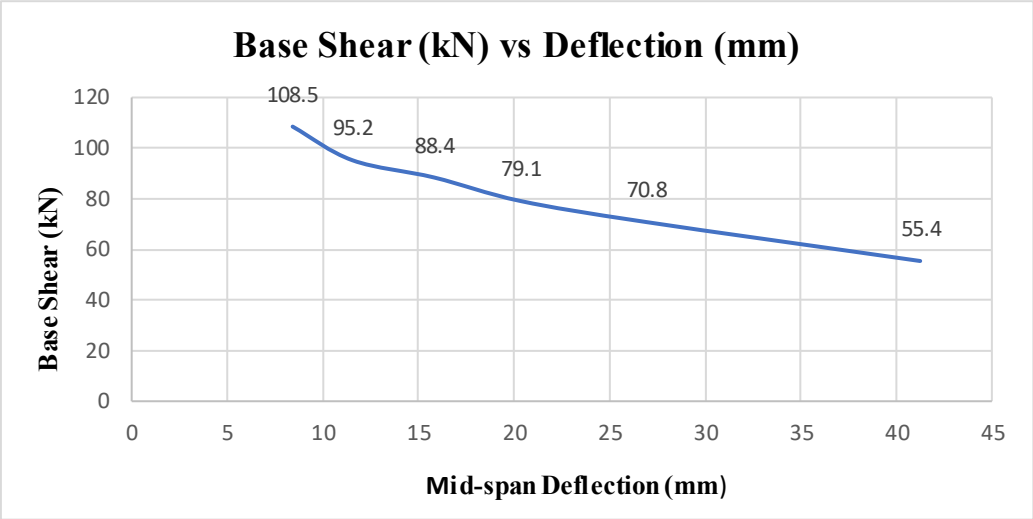


Figure 23:Base Shear (kN) vs Deflection (mm)

The Base Shear vs. Mid-span Deflection graph reveals an inverse correlation where increased deflection results in a steady decline in lateral force capacity. This trend confirms that higher beam slenderness leads to greater flexibility, compromising the structure's ability to attract and resist seismic base shear. The non-linear curve effectively maps the transition from high-stiffness behavior to a displacement-dominated response across different L/d ratios. These findings define a critical safety envelope for balancing structural strength against allowable seismic drift and deformation limits. Ultimately, the results prove that geometric proportions are the decisive factor in governing the beam's overall seismic performance.

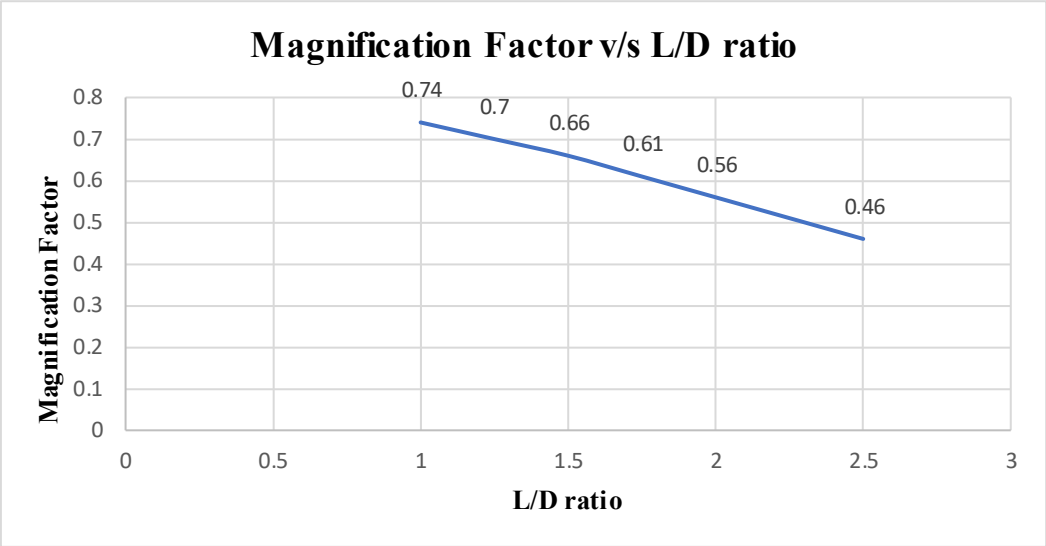
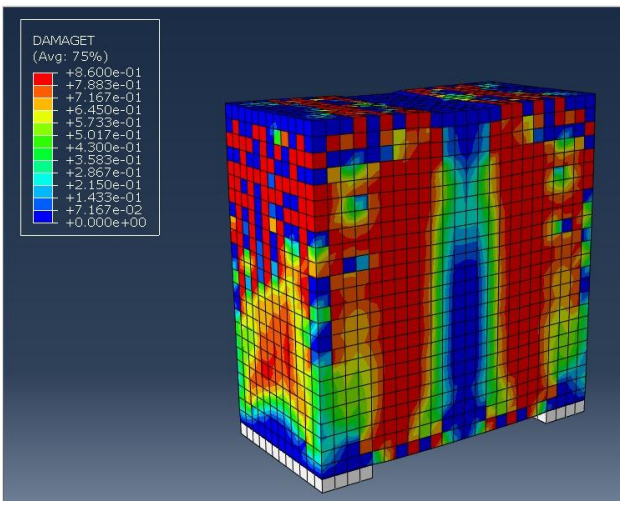
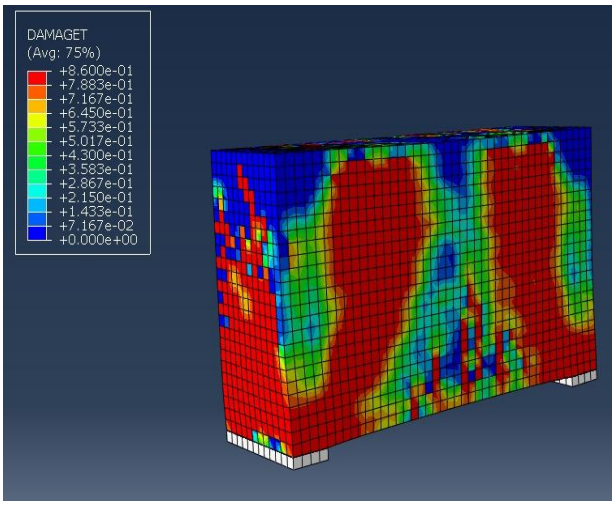


Figure 24:Magnification Factor v/s L/D ratio

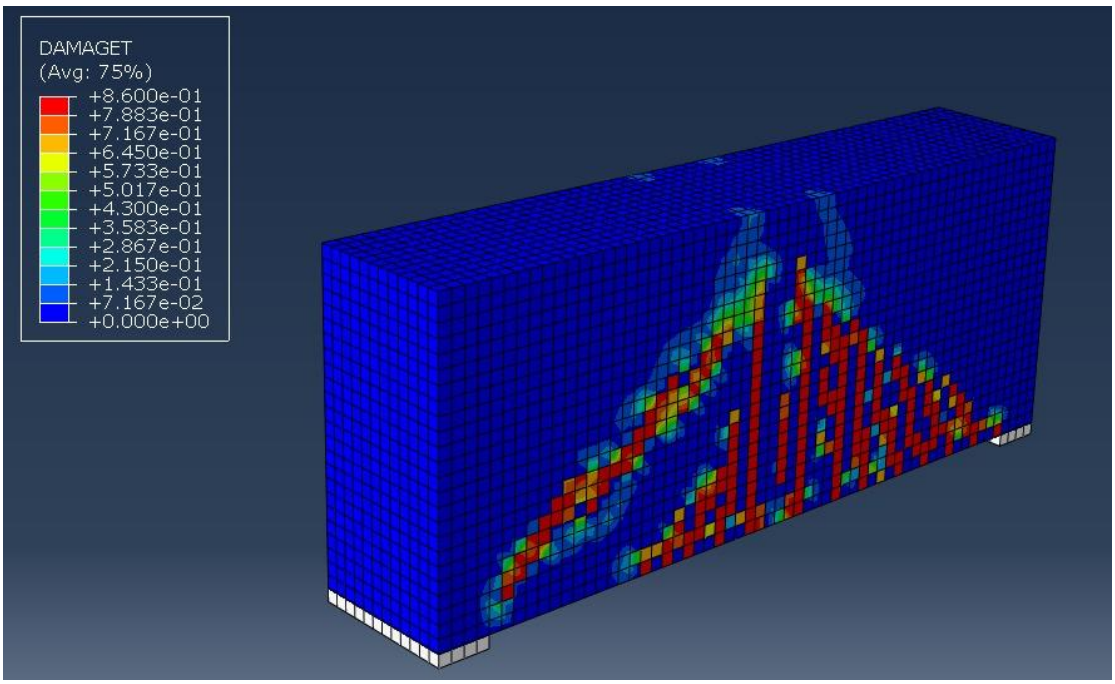
The Magnification Factor vs. L/D ratio graph indicates a sharp decline in amplification as the beam slenderness increases, starting from a peak of approximately 2.7 at a ratio of 1.0. This significant drop confirms that stiffer, short-span beams experience much higher dynamic amplification of ground motion compared to more flexible, longer spans. A localized stabilization or slight fluctuation is observed between ratios 1.25 and 1.5 before the factor continues its gradual downward trend toward 0.5. These results are vital for seismic design, as they demonstrate how increasing the L/D ratio can effectively move a structure away from critical resonance zones. Ultimately, the data provides a clear quantitative limit for controlling structural vibrations and ensuring stability during seismic events.



l/d = 1



l/d = 1.75



l/d = 2.5

Figure 25:Transitioning from Strut-and-Tie to Flexural Action for Ratios 1.0, 1.75, and 2.5

The DAMAGET contour images provide a visual map of the transition between different load-transfer mechanisms as the beam geometry changes. In the L/d = 1.0 model, the concentrated red zones along a diagonal path clearly illustrate the formation of Strut-and-Tie and Arch action, where forces travel directly from the load point to the supports. As the ratio increases, these patterns shift toward the bottom of the span, signaling a transition to a flexural response characterized by longitudinal cracking. The color intensity confirms that damage is most severe in the stiffest deep beams, which attract higher seismic force. Ultimately, these visualizations validate that structural slenderness dictates whether a beam fails through shear-driven arching or standard bending mechanisms.

7. CONCLUSION

The present study focused on the behaviour of reinforced concrete deep beams subjected to concentrated point loading, with particular emphasis on the influence of span-to-depth (L/D) ratio. Due to the presence of discontinuity regions, deep beams do not follow conventional flexural behaviour and instead transfer loads primarily through arching action. The use of ACI 318 provisions, along with finite element analysis in ABAQUS, enabled a detailed evaluation of structural response under both static and seismic loading conditions.

From the static analysis, it is observed that beams with lower L/D ratios exhibit higher load-carrying capacity and reduced deflection. For instance, the beam with **L/D=1.0** shows an ultimate load of **1182.25 kN** with a peak deflection of **0.250 mm**, whereas for **L/D=2.5**, the ultimate load reduces to **667.70 kN** and the deflection increases to **1.280 mm**. This clearly indicates a reduction in strength and stiffness with increasing L/D ratio, along with increased deformation, representing a transition from arch-dominated behaviour to a distributed load transfer mechanism.

The rebar stress distribution obtained from ABAQUS further supports these findings. The stress for lower L/D ratios reaches approximately **215.4 MPa** with a concentrated diagonal stress path, while for higher L/D ratios, the peak stress increases to about **424.3 MPa** and becomes more diffused across the beam depth. Higher slenderness ratios result in substantially higher strain for progressively smaller stress increments.

The seismic analysis also highlights the influence of geometry on dynamic response. For L/D=1.0, the base shear is approximately **95.2 kN** and peak acceleration is **7259 mm/s²**, whereas for L/D=2.5, the base shear reduces to **55.4 kN** and the peak acceleration reduces to **4512 mm/s²**.Transverse deflection rises from **8.42 mm** to **41.25 mm**. This indicates that lower L/D ratios provide higher stiffness and resistance, while higher ratios exhibit increased flexibility under seismic loading.

Overall, the study confirms that the span-to-depth ratio is a key parameter governing the behaviour and performance of deep beams under both static and dynamic conditions.

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APPENDIX

- ## Manual Calculations (ACI method)

For Point Load = 300 kN

a. For L/D = 1

L= 600 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Load due to self weight

w = 4.5*0.6 = 2.7 kN

Total load on beam including self weight, w_t = 300 + 2.7 = 302.7 kN

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 151.35 kN

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/300$

$\theta = 63.4°$

v. Strut Force, $C = R/\sin\theta = 151.35/0.894 = 169.3$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 169.3 * 0.447 = 75.675$ kN

vii. Area of Steel Required

$A_{st} = T/0.87fy = (75.675*10^3)/(0.87*500) = 173.96$ mm^2

viii. Provide Reinforcement

Provide 4 bars of 8 mm

Area of one bar = 113.1 mm^2

Total A_{st} = 201.06 mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , $f_{ce} = 0.85 * \beta_s f'_c$

$f_{ce} = 0.85 * 0.6*30 = 15.3$ MPa (For bottle-shaped strut, $\beta_s = 0.6$)

Assume strut width, W_s = 200 mm

Strut area $A_c = 300*200 = 60000$ mm^2

Strut capacity, $P_n = f_{ce} A_c = 918$ kN

Applied, C = 169.3 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

(According to ACI 318-19 Section 9.9.3.1)

Minimum distributed reinforcement ratio, $\rho_{min} = 0.0025$

Required reinforcement per unit length:

$A_{req} = \rho_{min} * b_w = 0.0025*300 = 0.75\ mm^2/mm = 750\ mm^2$

$750\ mm^2$ reinforcement per m

Use 8 mm diameter bars

Area of one bar, $A_{bar} = \pi d^2/4 = \pi 8^2/4 = 50.3$ mm^2

Spacing of Bars (ACI distributed reinforcement relation)

$A_{bar}/s \geq A_{req} \rightarrow 50.3/s \geq 0.75$

s = 67 mm

- ACI Maximum Spacing Check

(From ACI 318-19 Section 9.9.3.3)

$s \leq (d/5) \rightarrow s \leq 550/5 \rightarrow s \leq 110$ mm

Since 67 < 110 . Spacing OK.

Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

n = 600/65 = 9.23

Number of bars = spaces + 1, $N_v = 10.23$

Provide 11 bars

- Number of Horizontal Bars

n = 600/65 = 9.23

Number of bars = spaces + 1, $N_h = 10.23$

Provide 11 bars

Bottom tie reinforcement, 4 – 8 mm bars

Vertical web reinforcement,11- 8 mm @ 65 mm

Horizontal web reinforcement,11 - 8 mm @ 65 mm

b. <u>For L/D = 1.25</u>

L= 750 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Load due to self weight

w = 4.5*0.75 =3.375 kN

Total load on beam including self weight , w_t = 300 + 3.375 = 303.375 kN

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 303.4/2 = 151.7 kN

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/375$

$\theta = 58°$

v. Strut Force, $C = R/\sin\theta = 151.7/\sin(58°) = 178.88$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 178.88 * \cos(58°) = 94.79$ kN

vii. Area of Steel Required

$A_{st} = T/0.87fy = (94.79*10^3)/(0.87*500) = 217.91 mm^2$

viii. Provide Reinforcement

Provide 3 bars of 10 mm

Area of one bar = 78.5 mm^2

Total A_{st} = 235.5 mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , $f_{ce} = 0.85 * \beta_s f'_c$

$f_{ce} = 0.85 * 0.6*30 = 15.3$ MPa (For bottle-shaped strut, $\beta_s = 0.6$)

Assume strut width, $W_s = 200$ mm

Strut area $A_c = 300*200 = 60000 mm^2$

Strut capacity, $P_n = f_{ce} A_c = 918$ kN

Applied, C = 296.8 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

(According to ACI 318-19 Section 9.9.3.1)

Minimum distributed reinforcement ratio, $\rho_{min} = 0.0025$

Required reinforcement per unit length:

$A_{req} = \rho_{min} * b_w = 0.0025*300 = 0.75 mm^2/mm = 750 mm^2$

750 mm^2 reinforcement per m

Use 8 mm diameter bars

Area of one bar, $A_{bar} = \pi d^2/4 = \pi 8^2/4 = 50.3 mm^2$

Spacing of Bars (ACI distributed reinforcement relation)

$A_{bar}/s \geq A_{req} \rightarrow 50.3/s \geq 0.75$

s = 67 mm

- ACI Maximum Spacing Check

(From ACI 318-19 Section 9.9.3.3)

$s \leq (d/5) \rightarrow s \leq 550/5 \rightarrow s \leq 110$ mm

Since 67 < 110 . Spacing OK.

Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

n = 750/65 = 11.5

Number of bars = spaces + 1, $N_v = 12.5$

Provide 13 bars

- Number of Horizontal Bars

n = 600/65 = 9.23

Number of bars = spaces + 1, $N_h = 10.23$

Provide 11 bars

Bottom tie reinforcement, 3 – 10 mm bars

Vertical web reinforcement,13- 8 mm @ 65 mm

Horizontal web reinforcement,11 - 8 mm @ 65 mm

c. <u>For L/D = 1.5</u>

L= 900 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Load due to self weight

w = 4.5*0.9 = 4.05 kN

Total load on beam including self weight , w_t = 300 + 4.05 = 304.05 kN/m

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 304.05/2 = 152.025 kN

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/450$

$\theta = 53.13°$

v. Strut Force, $C = R/\sin\theta = 152.025/\sin(53.13°) = 190.03$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 190.03 * \cos(53.13°) = 114.02$ kN

vii. Area of Steel Required

A_{st} = T/0.87fy = (114.02*10^3)/(0.87*500) = 262.11 mm^2

viii. Provide Reinforcement

Provide 6 bars of 8 mm

Area of one bar = 50.3 mm^2

Total A_{st} = 301.8 mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , f_{ce}= 0.85* β_s f'_c

f_{ce}= 0.85* 0.6*30 = 15.3 MPa (For bottle-shaped strut, β_s = 0.6)

Assume strut width, W_s = 200 mm

Strut area A_c= 300*200 = 60000 mm^2

Strut capacity, P_n = f_{ce} A_c = 918 kN

Applied, C = 190.03 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

(According to ACI 318-19 Section 9.9.3.1)

Minimum distributed reinforcement ratio, ρ_{min} = 0.0025

Required reinforcement per unit length:

A_{req} = ρ_{min}*b_w = 0.0025*300 = 0.75 mm^2 /mm = 750 mm^2

750 mm^2 reinforcement per m

Use 8 mm diameter bars

Area of one bar, A_{bar} = $\pi d^2/4$ = $\pi 8^2/4$ = 50.3 mm^2

Spacing of Bars (ACI distributed reinforcement relation)

$A_{bar}/s \geq A_{req}$ → $50.3/s \geq 0.75$

s = 67 mm

- ACI Maximum Spacing Check

(From ACI 318-19 Section 9.9.3.3)

$s \leq (d/5)$ → $s \leq 550/5$ → $s \leq 110$ mm

Since 67 < 110 . Spacing OK.

Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

n = 900/65 = 13.85

Number of bars = spaces + 1, $N_v = 14.85$

Provide 15 bars

- Number of Horizontal Bars

n = 600/65 = 9.23

Number of bars = spaces + 1, $N_h = 10.23$

Provide 11 bars

Bottom tie reinforcement, 6 – 8 mm bars

Vertical web reinforcement,15- 8 mm @ 65 mm

Horizontal web reinforcement,11 - 8 mm @ 65 mm

d. <u>For L/D = 1.75</u>

L= 1050 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Load due to selfweight

w = 4.5*1.75 = 4.725 kN

Total load on beam including self weight, w_t = 300 + 4.725 = 304.725 kN

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 304.725/2 = 152.36 k

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/525$

$\theta = 48.81°$

v. Strut Force, $C = R/\sin\theta = 152.36/\sin(48.81°) = 202.46$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 202.46 * \cos(48.81°) = 133.33$ kN

vii. Area of Steel Required

$A_{st} = T/0.87fy = (133.33*10^3)/(0.87*500) = 306.5$ mm^2

viii. Provide Reinforcement

Provide 4 bars of 10 mm

Area of one bar = 78.5 mm^2

Total $A_{st} = 314$ mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , $f_{ce} = 0.85 * \beta_s f'_c$

$f_{ce} = 0.85 * 0.6*30 = 15.3$ MPa (For bottle-shaped strut, $\beta_s = 0.6$)

Assume strut width, W_s = 200 mm

Strut area A_c= 300*200 = 60000 mm^2

Strut capacity, $P_n = f_{ce} A_c$ = 918 kN

Applied, C = 202.46 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

  (According to ACI 318-19 Section 9.9.3.1)

  Minimum distributed reinforcement ratio, ρ_{min} = 0.0025

  Required reinforcement per unit length:

  $A_{req} = \rho_{min} * b_w$ = 0.0025*300 = 0.75 mm^2 /mm = 750 mm^2

  750 mm^2 reinforcement per m

  Use 8mm diameter bars

  Area of one bar, $A_{bar} = \pi d^2/4 = \pi 8^2/4$ = 50.3 mm^2

  Spacing of Bars (ACI distributed reinforcement relation)

  $A_{bar}/s \geq A_{req} \rightarrow 50.3/s \geq 0.75$

  s = 67 mm

- ACI Maximum Spacing Check

  (From ACI 318-19 Section 9.9.3.3)

  $s \leq (d/5) \rightarrow s \leq 550/5 \rightarrow s \leq 110$ mm

  Since 67 < 110 . Spacing OK.

  Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

  n = 1050/65 = 16.15

  Number of bars = spaces + 1, N_v = 17.15

  Provide 18 bars

- Number of Horizontal Bars

  n = 600/65 = 9.23

  Number of bars = spaces + 1, N_h = 10.23

  Provide 11 bars

Bottom tie reinforcement, 4 – 10 mm bars

Vertical web reinforcement,18- 8 mm @ 65 mm or

Horizontal web reinforcement,11 - 8 mm @ 65 mm

e. <u>For L/D = 2</u>

L= 1200 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Total load due to selfweight

w = 4.5*1.2 = 5.4 kN

Total load on beam including self weight, wt = 300 + 5.4 = 305.4 kN

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 305.4/2 = 152.7 kN

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/600$

$\theta = 45°$

v. Strut Force, $C = R/\sin\theta = 152.7/\sin(45°) = 215.95$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 215.95 * \cos(45°) = 152.7$ kN

vii. Area of Steel Required

$A_{st} = T/0.87fy = (152.7*10^3)/(0.87*500) = 351.03$ mm^2

viii. Provide Reinforcement

Provide 5 bars of 10 mm

Area of one bar = 78.5 mm^2

Total A_{st} = 392.5 mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , $f_{ce} = 0.85 * \beta_s f'_c$

$f_{ce} = 0.85 * 0.6*30 = 15.3$ MPa (For bottle-shaped strut, $\beta_s = 0.6$)

Assume strut width, $W_s = 200$ mm

Strut area $A_c = 300*200 = 60000$ mm^2

Strut capacity, $P_n = f_{ce} A_c = 918$ kN

Applied, C = 215.95 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

(According to ACI 318-19 Section 9.9.3.1)

Minimum distributed reinforcement ratio, $\rho_{min} = 0.0025$

Required reinforcement per unit length:

$A_{req} = \rho_{min} * b_w = 0.0025*300 = 0.75$ mm^2/mm = 750 mm^2

750 mm^2 reinforcement per m

Use 8 mm diameter bars

Area of one bar, $A_{bar} = \pi d^2/4 = \pi 8^2/4 = 50.3$ mm^2

Spacing of Bars (ACI distributed reinforcement relation)

$A_{bar}/s \geq A_{req} \rightarrow 50.3/s \geq 0.75$

s = 67 mm

- ACI Maximum Spacing Check

  (From ACI 318-19 Section 9.9.3.3)

  $s \leq (d/5) \rightarrow s \leq 550/5 \rightarrow s \leq 110$ mm

  Since 67 < 110 . Spacing OK.

  Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

  n = 1200/65 = 18.46

  Number of bars = spaces + 1, $N_v = 19.46$

  Provide 20 bars

- Number of Horizontal Bars

  n = 600/65 = 9.23

  Number of bars = spaces + 1, $N_h = 10.23$

  Provide 11 bars

Bottom tie reinforcement, 5 – 10 mm bars

Vertical web reinforcement,20- 8 mm @ 65 mm or

Horizontal web reinforcement,11 - 8 mm @ 65 mm

f. <u>For L/D = 2.5</u>

L= 1500 mm

i. Self Weight of Beam = 25*300*600 = 4.5 kN/m

ii. Total load due to selfweight

w = 4.5*1.5 = 6.75 kN

Total load on beam including self weight, w_t = 300 + 6.75 = 306.75 kN

iii. Support Reactions

For simply supported beam, R_A = R_B = W/2 = 306.75/2 = 153.375 kN

iv. Geometry of Deep Beam

Strut angle, $\tan\theta = 600/750$

$\theta = 38.66°$

v. Strut Force, $C = R/\sin\theta = 153.4/\sin(38.66°) = 245.56$ kN

vi. Tie Force (Bottom Reinforcement)

$T = C\cos\theta = 245.56 * \cos(38.66°) = 191.75$ kN

vii. Area of Steel Required

$A_{st} = T/0.87fy = (191.75*10^3)/(0.87*500) = 440.8$ mm^2

viii. Provide Reinforcement

Provide 4 bars of 12 mm

Area of one bar = 113.1 mm^2

Total A_{st} = 452.4 mm^2 ✓ Safe

ix. Check Strut Capacity (ACI 318)

Effective compressive strength , $f_{ce} = 0.85 * \beta_s f'_c$

$f_{ce} = 0.85 * 0.6*30 = 15.3$ MPa (For bottle-shaped strut, $\beta_s = 0.6$)

Assume strut width, $W_s = 200$ mm

Strut area $A_c = 300*200 = 60000$ mm^2

Strut capacity, $P_n = f_{ce} A_c = 918$ kN

Applied, C = 245.56 kN ✓ Safe

x. Final Reinforcement (ACI recommendation for deep beams)

- Web Reinforcement Design ,

  (According to ACI 318-19 Section 9.9.3.1)

  Minimum distributed reinforcement ratio, $\rho_{min} = 0.0025$

  Required reinforcement per unit length:

  $A_{req} = \rho_{min} * b_w = 0.0025*300 = 0.75$ mm^2/mm = 750 mm^2

  750 mm^2 reinforcement per m

  Use 8 mm diameter bars

  Area of one bar, $A_{bar} = \pi d^2/4 = \pi 8^2/4 = 50.3$ mm^2

  Spacing of Bars (ACI distributed reinforcement relation)

  $A_{bar}/s \geq A_{req} \rightarrow 50.3/s \geq 0.75$

  $s = 67$ mm

- ACI Maximum Spacing Check

  (From ACI 318-19 Section 9.9.3.3)

  $s \leq (d/5) \rightarrow s \leq 550/5 \rightarrow s \leq 110$ mm

Since 67 < 110 . Spacing OK.

Adopt practical spacing, s = 65 mm

- Number of Vertical Bars

  n = 1500/65 = 23.07

  Number of bars = spaces + 1, N_v = 24.07

  Provide 25 bars

- Number of Horizontal Bars

  n = 600/65 = 9.23

  Number of bars = spaces + 1, N_h = 10.23

Provide 11 bars

Bottom tie reinforcement, 4 – 12 mm bars

Vertical web reinforcement,25- 8 mm @ 65 mm or

Horizontal web reinforcement,11 - 8 mm @ 65 mm

L/d ratio	Strut Angle	Strut Force (kN)	Tie Force (kN)	Area of steel (mm^2)	Bottom Reinforcement	Verticl Reinforcement	Horizontal Reinforcement
1	63.4	169.3	75.675	173.96	4 – 8 mm bars	11- 8 mm @ 65 mm	11 - 8 mm @ 65 mm
1.25	58	178.88	94.79	217.91	3 – 10 mm bars	13- 8 mm @ 65 mm	11 - 8 mm @ 65 mm
1.5	53.13	190.03	114.02	262.11	6 – 8 mm bars	15- 8 mm @ 65 mm	11 - 8 mm @ 65 mm
1.75	48.81	202.46	133.33	306.5	4 – 10 mm bars	18- 8 mm @ 65 mm	11 - 8 mm @ 65 mm
2	45	215.95	152.7	351.03	5 – 10 mm bars	20- 8 mm @ 65 mm	11 - 8 mm @ 65 mm
2.5	38.66	245.56	191.75	440.8	4 – 12 mm bars	25- 8 mm @ 65 mm	11 - 8 mm @ 65